Chapter # 02
Blood Circulatory System of Man

Student Learning Outcomes

After studying this chapter, students will be able to:

- [B-12-R-36] State the location of heart in the body and define the role of pericardium.
- [B-12-R-37] Describe the structure of the walls of heart and rationalize the thickness of the walls of each chamber.
- [B-12-R-38] Trace the flow of blood through the heart as regulated by the valve.
- [B-12-R-39] State the phases of heartbeat.
- [B-12-R-40] Explain the roles of A node, AV node and Purkinje fibers in controlling the heartbeat.
- [B-12-R-41] List the principles and uses of Electrocardiogram.
- [B-12-R-42] Describe the detailed structure of arteries, veins and capillaries.
- [B-12-R-43] Describe the role of arterioles in vasoconstriction and vasodilation.
- [B-12-R-44] Describe the role of pre-capillary sphincters in regulating the flow of blood through the capillaries.
- [B-12-R-45] Trace the path of the blood through the pulmonary and systemic circulation (coronary, hepatic-portal and renal circulation).
- [B-12-R-46] Compare the rate of blood flow through arteries, arterioles capillaries, venules and veins.
- [B-12-R-47] Define blood pressure and explain its periods of systolic and diastolic pressure.
- [B-12-R-48] State the role of baroreceptors and volume receptor in regulating the blood pressure.
- [B-12-R-49] Define the term thrombus and differentiate between thrombus and embolus.
- [B-12-R-50] Identify the factors causing atherosclerosis and arteriosclerosis.
- [B-12-R-51] Categorize Angina pectoris, heart attack, and heart failure as the stages of cardiovascular disease development.
- [B-12-R-52] State the congenital heart problem related to the malfunction of cardiac valves.
- [B-12-R-53] Describe the principles of angiography.
- [B-12-R-54] Outline the main principle of coronary bypass, angioplasty and open-heart surgery.
- [B-12-R-55] Define hypertension and describe the factor that regulate blood pressure and lead to hypertension and hypotension.
- [B-12-R-56] List the changes in lifestyles that can protect man from hypertension and cardiac problems.
- [B-12-R-57] Describe the formation, composition and function of intercellular fluid.
- [B-12-R-58] Compare the composition of intercellular fluid with that of lymph.
- [B-12-R-59] State the structure and role of lymph capillaries, lymph vessels and lymph trunks.
- [B-12-R-60] Describe the functions of lymph nodes and state the role of spleen as containing lymphoid tissue.

You have already learned in earlier chapters that digestion is the process by which large, insoluble food molecules are broken down into smaller, soluble ones, and that all living organisms are made of cells. Each cell in the body requires a constant supply of nutrients from the small intestine and oxygen from the lungs to sustain life processes. Likewise, waste materials such as carbon dioxide and nitrogenous compounds must be efficiently removed via the lungs and kidneys. Because the human body is too large for simple diffusion to meet these demands, an internal transport system is essential. The circulatory system fulfills this role by transporting oxygen and carbon dioxide, delivering nutrients to cells, and carrying metabolic waste to designated sites for excretion. The cardiovascular system comprises a muscular heart, three types of blood vessels—arteries, capillaries, and veins—and the circulating blood. The scientific study of cardiovascular diseases is known as **angiology**.

SLO [B-12-R-36]
State the location of heart in the body and define the role of pericardium.

LOCATION HUMAN HEART AND ROLE OF PERICARDIUM

Location of the Heart

The human heart is a **muscular, cone-shaped organ** located in the **thoracic cavity** between the lungs, in a space known as the **mediastinum**.

- It lies **slightly left of the midline** of the body, with its broad upper end (the **base**) directed upward and to the right, and its pointed lower end (the **apex**) pointing downward, forward, and to the left.
- The apex rests on the **diaphragm**, while the base lies beneath the second rib.
- Approximately two-thirds of the heart's mass lies to the left of the mid-sagittal plane, and one-third lies to the right.

This central position ensures that the heart can efficiently pump blood to both the **systemic and pulmonary** circulations.

Role of the Pericardium

The **pericardium is a double-walled membranous sac** that completely encloses the heart, serving as a **protective shield, supportive sling, and natural lubricant** for this vital organ. It ensures that the heart remains securely positioned in the thoracic cavity and can contract continuously without damage or interference from surrounding tissues.

1. **Fibrous Pericardium – The Outer Protective Layer**
   - This is the **tough, inelastic outer covering** composed of dense connective tissue.
   - **Anchorage:** Firmly attaches the heart to surrounding structures such as the **diaphragm** (below), **sternum** (in front), and great blood vessels (above). This prevents displacement of the heart during vigorous movements or changes in body position.
   - **Protection:** Acts as a sturdy barrier against mechanical injury from nearby organs.
   - **Overfilling Prevention:** Its non-stretchable nature stops the heart from **over-distending** when blood volume rises suddenly, thus maintaining structural integrity and preventing inefficient pumping.
2. **Serous Pericardium – The Inner Fluid-Lined Layer**

   Inside the fibrous pericardium lies the **serous pericardium**, a **delicate, double-layered membrane** designed for frictionless movement. It has two continuous layers:
   - **Parietal Layer:** Lines the inner surface of the fibrous pericardium, acting as a smooth lining.
   - **Visceral Layer (Epicardium):** Adheres tightly to the outer surface of the heart, forming the external layer of the heart wall itself.

   The Pericardial Cavity and Serous Fluid
   - Between the parietal and visceral layers is a **narrow space called the pericardial cavity**.
   - This cavity contains **a thin film of serous fluid**—a slippery secretion that acts as a **biological lubricant**.
   - Functionally, this fluid **minimises friction** between the two layers during each heartbeat, ensuring **smooth, uninterrupted cardiac movements** even at high pumping rates.

SLO [B-12-R-37]
Describe the structure of the walls of heart and rationalize the thickness of the walls of each chamber.

STRUCTURE OF THE WALLS OF THE HEART AND THEIR FUNCTIONAL RATIONALE

Structure of the Heart

The heart is divided into four chambers: two upper **atria** (from Latin *atrium*, meaning "entrance hall") and two lower **ventricles** (from Latin *venter*, meaning "belly"). The atria are located above the ventricles and serve as receiving chambers, while the ventricles function as powerful pumping chambers.

The **heart wall** consists of three distinct layers. The **epicardium** is a thin serous membrane forming the smooth outer surface of the heart. Beneath it lies the **myocardium**, the thick muscular middle layer composed of cardiac muscle cells responsible for contraction. The innermost layer, the **endocardium**, is a smooth lining of simple squamous epithelium over connective tissue, forming the internal surface of the heart chambers. The **heart valves** arise as folds of the endocardium, forming double layers with connective tissue between them.

The thickness of each chamber wall reflects its function. The **atria** have relatively thin walls, as they only need to pump blood into the adjacent ventricles. In contrast, the **ventricles** have thicker walls to generate sufficient force to expel blood from the heart. The **left ventricle** has the thickest wall, as it must pump blood throughout the systemic circulation. The **right ventricle**, with a wall approximately one-third as thick, pumps blood to the nearby lungs.

The **right atrium** receives deoxygenated blood via the **superior vena cava**, **inferior vena cava**, and **coronary sinus**, while the **left atrium** receives oxygenated blood from the **four pulmonary veins**. The atria are separated by the **interatrial septum** and communicate with the ventricles through **atrioventricular canals**. The **right ventricle** opens into the **pulmonary trunk**, and the **left ventricle** opens into the **aorta**. The two ventricles are divided by the **interventricular septum**.

Each atrioventricular canal contains an **atrioventricular (AV) valve** made of flaps called **cusps**. The **tricuspid valve**, with three cusps, lies between the right atrium and right ventricle. The **bicuspid** or **mitral valve**, with two cusps, is located between the left atrium and left ventricle. Inside each ventricle are **papillary muscles**, cone-shaped projections that anchor **chordae tendineae**—strong connective tissue cords that attach to the valve cusps. During ventricular contraction, the papillary muscles contract, pulling on the chordae tendineae to prevent the valves from inverting into the atria.

At the exit of each ventricle are **semilunar valves**: the **pulmonary valve** in the pulmonary trunk and the **aortic valve** in the aorta. These valves ensure unidirectional flow of blood from the heart into the major arteries.

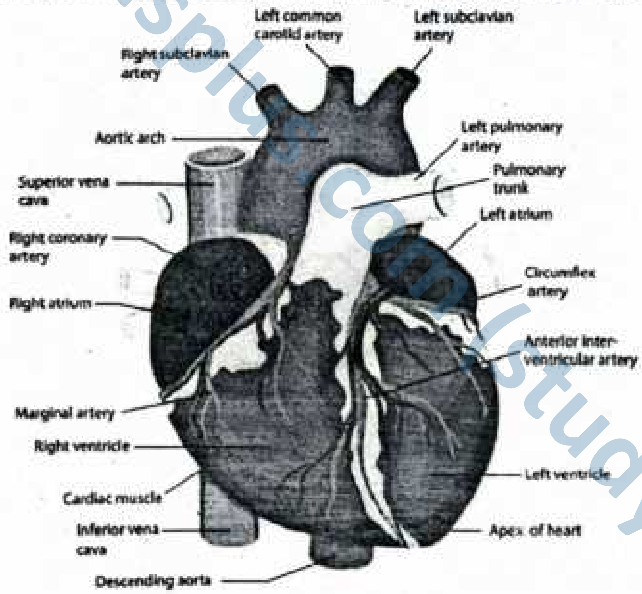


Figure: Human heart, external view

▸ **Comparison of structure and functions of four chambers of human heart**

Feature	Right Atrium	Right Ventricle	Left Atrium	Left Ventricle
Receives Blood From	Superior and inferior vena cava	Right atrium	Pulmonary veins	Left atrium
Type of Blood	Deoxygenated	Deoxygenated	Oxygenated	Oxygenated
Pumps Blood To	Right ventricle	Pulmonary artery	Left ventricle	Aorta
Wall Thickness	Thin	Thicker than atrium	Thin	Thickest of all chambers

Main Function	Collects deoxygenated blood from body	Pumps deoxygenated blood to lungs	Collects oxygenated blood from lungs	Pumps oxygenated blood to entire body
Valve Involved	Tricuspid valve (to right ventricle)	Pulmonary valve (to pulmonary artery)	Bicuspid/mitral valve (to left ventricle)	Aortic valve (to aorta)

- **Differences Among Heart Chambers**

Feature	Atria (Right & Left)	Ventricles (Right & Left)
Position	Upper chambers of the heart	Lower chambers of the heart
Wall Thickness	Thin walls	Thick walls
Function	Receive blood entering the heart	Pump blood out of the heart
Type of Blood	Right atrium: deoxygenated Left atrium: oxygenated	Right ventricle: deoxygenated Left ventricle: oxygenated
Connected Vessels	Right: vena cavae Left: pulmonary veins	Right: pulmonary artery Left: aorta

- **Differences Between Right Atrium and Left Atrium**

Feature	Right Atrium	Left Atrium
Blood Received From	Superior and inferior vena cava	Pulmonary veins
Type of Blood	Deoxygenated	Oxygenated
Function	Receives blood from body	Receives blood from lungs
Wall Structure	Thinner than left atrium	Slightly thicker than right atrium

- **Differences Between Right Ventricle and Left Ventricle**

Feature	Right Ventricle	Left Ventricle
Pumps Blood To	Lungs via pulmonary artery	Body via aorta
Type of Blood	Deoxygenated	Oxygenated
Wall Thickness	Thinner (short distance, low pressure)	Thickest (high pressure to reach whole body)
Function	Pumps blood to lungs	Pumps blood to entire body

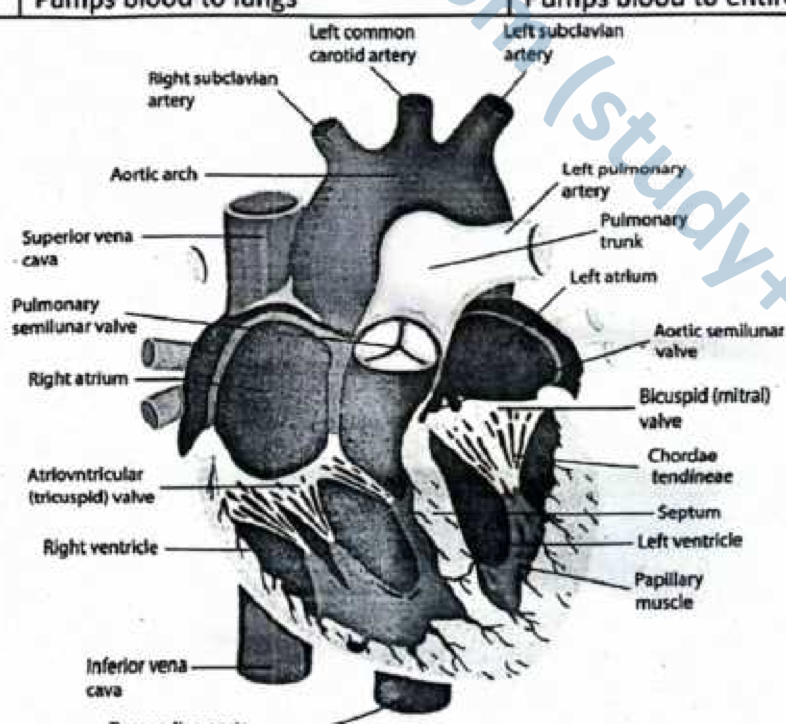


Figure: Dissection of a human heart, as seen from front, with the ventral part of both atria and both ventricles removed

Rationalizing the Thickness of Chamber Walls

The **myocardium**—the muscular middle layer of the heart wall—shows the greatest variation in thickness among the four chambers. This difference is not random; it reflects the **specific workload, distance, and pressure requirements** of each chamber. In essence, the more force and distance required to pump blood, the thicker the muscle wall.

- **Right Atrium – Thin-Walled and Low-Pressure Chamber**
  - **Role:** Acts as a receiving chamber for deoxygenated blood returning from the body via the superior and inferior venae cavae.
  - **Reason for Thinness:**
    - The right atrium only needs to push blood into the right ventricle, which lies immediately below it.
    - This short distance requires **minimal force**, so a thin myocardial layer is sufficient.
    - Excess muscle here would be wasteful and unnecessary for its low-pressure function.
- **Left Atrium – Thin-Walled but Oxygen-Rich**
  - **Role:** Receives oxygenated blood from the lungs through the pulmonary veins.
  - **Reason for Thinness:**
    - The left atrium pushes blood a short distance into the left ventricle.
    - Although this blood is rich in oxygen, the **energy demand for movement is low** because the pathway is short and requires little pressure.
    - Therefore, its myocardial thickness is similar to that of the right atrium.
- **Right Ventricle – Moderately Thick Wall for Pulmonary Circulation**
  - **Role:** Pumps deoxygenated blood to the lungs via the pulmonary artery.
  - **Reason for Moderate Thickness:**
    - The lungs are close to the heart, but some **force is required** to push blood through the pulmonary arteries and into the network of lung capillaries.
    - However, pressure must be kept **low** to avoid damaging the delicate alveolar capillaries.
    - As a result, the right ventricular wall is thicker than the atria but significantly thinner than the left ventricle.
- **Left Ventricle – Thickest Wall for Systemic Circulation**
  - **Role:** Pumps oxygenated blood into the aorta for distribution throughout the entire body.
  - **Reason for Extreme Thickness:**
    - Must generate **high pressure** to overcome systemic vascular resistance in arteries and reach tissues from head to toe.
    - Needs to sustain forceful contractions over a **longer distance** than any other chamber.
    - Its myocardium is **2–3 times thicker** than that of the right ventricle to meet these demands.

SLO [B-12-R-38]
Trace the flow of blood through the heart as regulated by the valve.

FLOW OF BLOOD THROUGH THE HEART (PATHWAY OF BLOOD THROUGH THE HEART)

The **superior** and **inferior vena cava**, both carrying deoxygenated blood from the body, deliver it into the **right atrium.** From there, blood passes through the **tricuspid valve** into the **right ventricle,** which then pumps it through the **pulmonary semilunar valve** into the **pulmonary trunk**. The pulmonary trunk divides into two **pulmonary arteries** that transport blood to the lungs for oxygenation.

Oxygenated blood returns from the lungs via **four pulmonary veins**, which empty into the **left atrium.** Blood then flows through the **bicuspid** (or **mitral**) valve into the **left ventricle.** The left ventricle pumps blood through the **aortic semilunar valve** into the **aorta**, which distributes it to the rest of the body.

The heart functions as a **double pump**: the **right side** sends deoxygenated blood to the **lungs,** while the **left side** delivers oxygenated blood to the **systemic circulation.** This separation ensures efficient gas exchange and nutrient distribution throughout the body.

▸ **Comparison the function of right side of the heart with the left side of the heart.**

Right Side of the Heart (De-oxygenated blood)	Left Side of the Heart (Oxygenated blood)
1. Right atrium receives deoxygenated blood from the body through the superior and inferior vena cava. ⬇	1. Left atrium receives oxygenated blood from the lungs via the pulmonary veins. ⬇
2. Tricuspid valve, located between right atrium and right ventricle, prevents backflow during ventricular contraction. ⬇	2. Bicuspid (mitral) valve, located between left atrium and left ventricle, prevents backflow during ventricular contraction. ⬇
3. Right ventricle pumps deoxygenated blood to the lungs through the pulmonary artery. ⬇	3. Left ventricle pumps oxygenated blood to the entire body via the aorta; it has the thickest muscular wall. ⬇
4. Pulmonary semilunar valve at the base of the pulmonary artery prevents backflow into the right ventricle.	4. Aortic semilunar valve at the base of the aorta prevents backflow into the left ventricle.

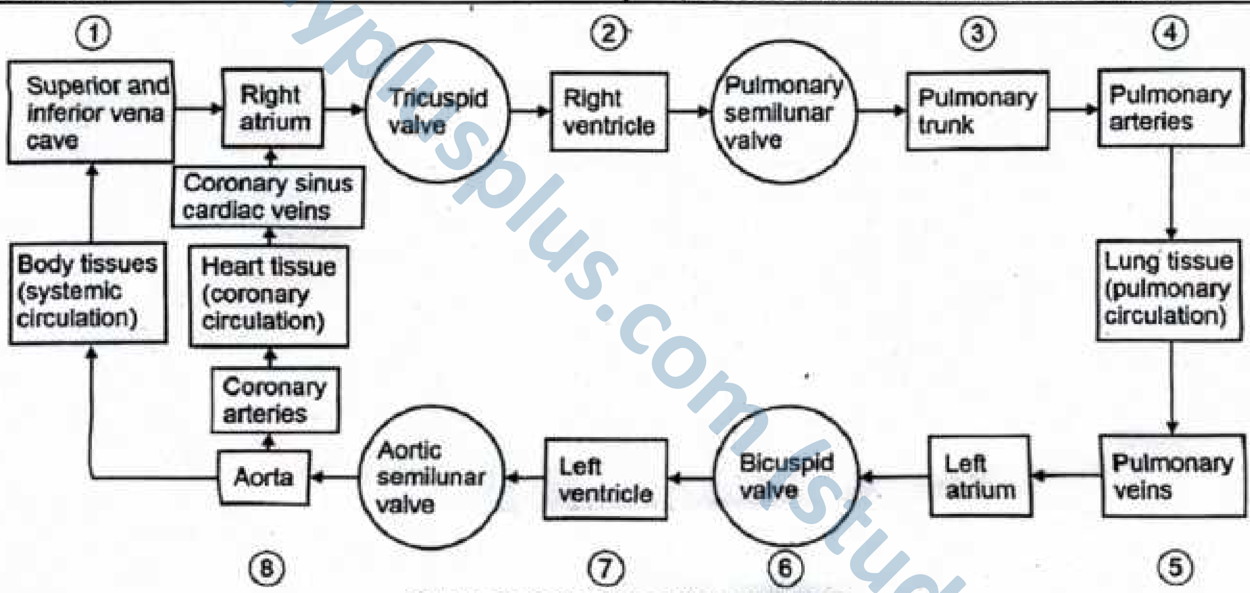


Figure: Passage of blood through heart

SLOs Based (MCQs)

HUMAN HEART

1. Which of the following structures delivers oxygenated blood to the left atrium?

A. Pulmonary trunk B. Superior vena cava
C. Four pulmonary veins D. Coronary sinus

2. Which valve prevents the backflow of blood from the left ventricle into the left atrium?

A. Aortic semilunar valve
B. Tricuspid valve
C. Pulmonary semilunar valve
D. Bicuspid (mitral) valve

3. The thick muscular layer of the heart wall responsible for contraction is the:

A. Endocardium B. Myocardium
C. Epicardium D. Pericardium

4. The right ventricle pumps blood into which of the following?

A. Aorta B. Pulmonary veins
C. Pulmonary trunk D. Coronary arteries

5. What is the function of the chordae tendineae in the heart?

A. Generate electrical impulses
B. Transport oxygen to the myocardium
C. Prevent inversion of valve cusps during ventricular contraction
D. Stimulate contraction of the atria

6. **Which of the following statements best explains why the left ventricular wall is thicker than the right?**
A. It receives oxygen-rich blood
B. It pumps blood a longer distance through the body
C. It contains pacemaker cells
D. It stores deoxygenated blood

7. **The pericardium serves all of the following functions EXCEPT:**
A. Lubricating the heart
B. Holding the heart in position
C. Acting as a muscular pump
D. Preventing over-expansion of the heart

Answer Key

1. C	2. D	3. B	4. C	5. C	6. B	7. C

Short Questions

Q.1 Why is the heart called a double pump?

Ans. The heart functions as a double pump because it drives two separate circulations simultaneously. The right ventricle pumps deoxygenated blood to the lungs via the pulmonary arteries for oxygenation. The left ventricle pumps oxygenated blood to the entire body through the aorta. This arrangement ensures continuous gas exchange in the lungs and nutrient/oxygen delivery to body tissues.

MARKING SCHEME (3 Marks)
- Role of right ventricle – 1.5 marks
- Role of left ventricle – 1.5 marks

Q.2 Describe the location and orientation of the human heart.

Ans. The heart is located in the mediastinum of the thoracic cavity, flanked by the lungs, with the vertebral column posteriorly and the sternum anteriorly. Its base lies beneath the second rib, and its apex points downward, forward, and to the left, reaching the fifth intercostal space. This position places the apex closer to the left side of the chest.

MARKING SCHEME (3 Marks)
- Location – 1.5 marks
- Orientation – 1.5 marks

Q.3 Name the three layers of the heart wall and state their functions.

Ans.
- **Epicardium:** Smooth, outer serous membrane that protects the heart surface.
- **Myocardium:** Thick, muscular middle layer responsible for pumping action.
- **Endocardium:** Thin inner lining of chambers and valves, ensuring smooth blood flow and preventing clot formation.

MARKING SCHEME (3 Marks)
- Epicardium – 1 mark
- Myocardium – 1 mark
- Endocardium – 1 mark

Q.4 What is the role of the pericardium?

Ans. The pericardium is a double-layered sac surrounding the heart. It prevents overexpansion during filling, reduces friction with serous fluid during heartbeats, and anchors the heart in position within the thoracic cavity. It consists of an inner visceral layer and an outer parietal layer forming the pericardial cavity.

MARKING SCHEME (3 Marks)
- Prevents overexpansion – 1 mark
- Reduces friction – 1 mark
- Holds heart in place – 1 mark

Q.5 Compare the thickness of the atrial and ventricular walls.

Ans. Atrial walls are thin because they only pump blood a short distance into adjacent ventricles, requiring little force. Ventricular walls are thicker, especially in the left ventricle, which must generate high pressure to pump blood throughout the body. The right ventricle is moderately thick as it pumps to the lungs only.

MARKING SCHEME (3 Marks)
- Thickness of atrial walls – 1.5 marks
- Thickness of ventricular walls – 1.5 marks

Q.6 How do papillary muscles and chordae tendineae work together?

Ans. Papillary muscles contract during ventricular systole, pulling on the chordae tendineae. This action holds the valve cusps in place, preventing backflow of blood into the atria and ensuring one-way circulation.

MARKING SCHEME (3 Marks)
- Function of papillary muscles – 1.5 marks
- Function of chordae tendineae – 1.5 marks

Q.7 Trace the flow of blood through the heart starting from the vena cava.

Ans. Deoxygenated blood enters the right atrium via the superior and inferior vena cava → passes through tricuspid valve to right ventricle → moves into pulmonary trunk via pulmonary valve → travels to lungs → oxygenated blood returns via pulmonary veins to left atrium → passes through bicuspid valve to left ventricle → pumped out via aorta to the body.

MARKING SCHEME (3 Marks)
- Flow of deoxygenated blood – 1.5 marks
- Flow of oxygenated blood – 1.5 marks

Q.8 Why does the tricuspid valve have three flaps and the bicuspid valve only two, despite higher pressure on the left side?

Ans. The tricuspid valve, on the right side, has three cusps to prevent backflow at low pressure when blood flows to the lungs. The bicuspid (mitral) valve, on the left side, has two cusps that are thicker and stronger to withstand the high pressure needed to pump blood to the body. Valve adaptation is based on cusp strength, not number.

MARKING SCHEME (3 Marks)
- Role of tricuspid valve – 1.5 marks
- Role of bicuspid valve – 1.5 marks

Q.9 What is the difference of pressure in right ventricle and left ventricle?

Ans. The right ventricle pumps blood at low pressure to nearby lungs, enabling efficient gas exchange with minimal resistance. The left ventricle pumps blood at high pressure throughout the body, overcoming greater vascular resistance.

MARKING SCHEME (3 Marks)
- Pressure in right ventricle with reason – 1.5 marks
- Pressure in left ventricle with reason – 1.5 marks

SLO [B-12-R-39]
State the phases of heartbeat.

PHASES OF HEARTBEAT

In a continuous, rhythmic cycle heart is passively filled with blood from the large veins and then the heart actively contracts, propelling the blood throughout the body. Its alternating relaxations and contractions make up the cardiac cycle. The cardiac cycle is a sequence of one heartbeat.

Phases of heartbeat

The heart functions in a continuous, rhythmic cycle in which it passively fills with blood from the major veins and then actively contracts to propel blood throughout the body. This coordinated sequence of alternating contraction and relaxation is known as the **cardiac cycle**, and it constitutes a single heartbeat.

▶ Phases of the Heartbeat

The term **systole** refers to contraction, and **diastole** refers to relaxation or dilation.

- **Atrial systole** is the contraction of the atrial myocardium, while **atrial diastole** refers to its relaxation.
- **Ventricular systole** and **diastoSLOle** represent contraction and relaxation of the ventricular myocardium, respectively.
- When the terms *systole* and *diastole* are used without specification, they generally refer to ventricular activity.

During **atrial diastole**, deoxygenated blood enters the right atrium from the venae cavae. Initially, the **tricuspid** and **bicuspid (mitral)** valves are closed. As the atria fill, pressure builds and exceeds that of the relaxed ventricles, causing the atrioventricular valves to open passively. In **atrial systole**, both atria contract simultaneously, pushing blood into the ventricles, which are still in diastole.

At this stage:

- *Atrioventricular valves* (bicuspid and tricuspid) are **open**
- *Semilunar valves* (aortic and pulmonary) are **closed**

Next, **ventricular systole** begins almost immediately. The ventricles contract, sharply increasing intraventricular pressure. This rise in pressure forces the atrioventricular valves to close, preventing backflow into the atria, and simultaneously opens the semilunar valves, allowing blood to be ejected into the **aorta** and **pulmonary artery**.

At this stage:

- Atrioventricular valves are **closed**
- Semilunar valves are **open**

During **ventricular diastole,** the pressure within the aorta and pulmonary artery exceeds that in the relaxing ventricles. This reverse pressure causes the semilunar valves to close, preventing the backflow of blood. Meanwhile, the atrioventricular valves reopen as atrial pressure builds once more, initiating the next cycle. At this stage:

- Atrioventricular valves are **open**
- Semilunar valves are **closed**

A complete cardiac cycle takes approximately **0.7 to 0.8 seconds**, depending on the contractile efficiency of the cardiac muscle. Between beats, the heart muscle rests for about **0.1 to 0.3 seconds**, ensuring sustained function without fatigue.

When a stethoscope is used to listen to the heart sounds, distinct sounds normally are heard. The first heart sound is a low-pitched sound, often described as a "lub" sound. It is caused by vibration of the atrioventricular valves which close near the beginning of ventricular systole. The second heart sound is a higher pitched sound often described as a "dub" sound. It results from closure of the aortic and pulmonary samilunar valves, near the end of systole. 'lub' is also written as 'lubb' and 'dub' as 'dupp'.

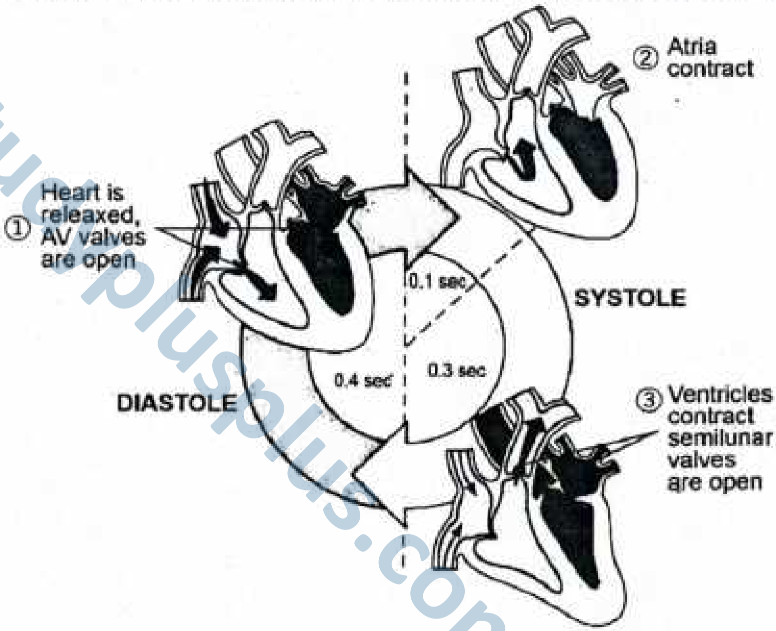


Figure: Cardiac cycle

SLO [B-12-R-40]
Explain the roles of A node, AV node and Purkinje fibers in controlling the heartbeat.

CONDUCTING SYSTEM OF THE HEART

The Myogenic Nature of the Heart

Unlike skeletal muscles, the heart can continue to beat even after being removed from the body—provided it is supplied with nutrients and oxygen. This is because **cardiac muscle is myogenic**, meaning its rhythmic contractions originate from within the muscle itself, rather than requiring stimulation from nerves. This self-generated rhythmicity is a defining feature of cardiac function.

Intrinsic Rhythmicity and the Conduction System

Cardiac muscle possesses **intrinsic rhythmicity**, which allows each heartbeat to arise and propagate without any **extrinsic neural input**. A specialized network of **interconnected cardiac muscle fibres**, responsible for initiating and conducting electrical impulses, ensures that the heart contracts in a coordinated manner. This network is known as the **conduction system of the heart**, and it regulates the **cardiac cycle**.

Components of the Conduction System

The conduction system is composed of:

1. Sinoatrial node (SA node)
2. Atrioventricular node (AV node)
3. Atrioventricular bundle (Bundle of His)
4. Purkinje fibres (conducting myofibrils)

1. Sinoatrial Node (SA Node) – The Pacemaker

The **SA node** is a small but vital group of specialized cardiac muscle fibres located in the **upper wall of the right atrium**, near the entry point of the **superior vena cava**. It originates from embryonic **sinus venosus tissue**, which is why it is called the *sinoatrial* node.

SA node cells generate **action potentials** at the highest frequency among cardiac cells, making the SA node the **natural pacemaker** of the heart. Under resting conditions, these impulses spread through the atrial muscle and reach the **AV node** in about **0.04 seconds**.

2. Atrioventricular Node (AV Node)

Located near the junction of the **right atrium and right ventricle**, the **AV node** serves as the only electrical connection between the atria and ventricles. Within the AV node, the conduction of impulses is **intentionally slow**, resulting in a delay of about **0.11 seconds.** This delay is essential, as it allows the atria to fully contract and empty their blood into the ventricles **before** the ventricles begin to contract.

3. Atrioventricular Bundle (Bundle of His)

The **AV bundle**, also called the **bundle of His**, arises from the AV node and passes through the **fibrous skeleton of the heart** into the **interventricular septum**. Within the septum, the bundle splits into the **right and left bundle branches**, which travel along the septum toward the apex of the heart, just beneath the **endocardium**.

4. Purkinje Fibres – Rapid Impulse Conductors

At the terminal ends of the bundle branches, the conduction fibres become **Purkinje fibres**—large-diameter cardiac muscle fibres that contain fewer myofibrils and do not contract forcefully. These fibres are interconnected by **intercalated discs** that contain abundant **gap junctions**, allowing action potentials to spread **very rapidly** throughout the ventricular myocardium.

This rapid conduction ensures **simultaneous contraction** of both ventricles, which is crucial for effective pumping.

Timing and Coordination in the Cardiac Cycle

From the SA node to the AV node, conduction takes approximately **0.04 seconds.** The **total delay** at the AV node is about **0.11 seconds,** bringing the total delay to **0.15 seconds** before the impulse reaches the ventricles. This carefully timed delay allows:

- Complete atrial contraction
- Full ventricular filling
- Efficient and coordinated ventricular systole

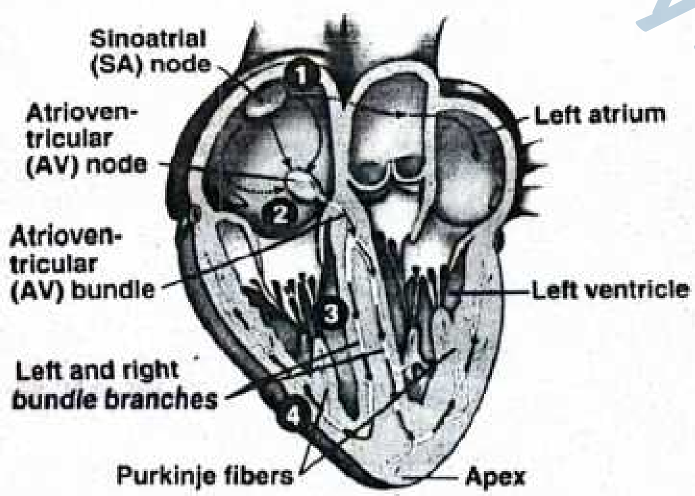


Figure: Conducting System of the Heart

1. **Action potentials originate in the sinoatrial (SA) node and travel across the wall of the atrium (arrows) from the SA node to the atrioventricular (AV) node.**
2. **Action potentials pass through the AV node and along the atrioventricular (AV) bundle, which extends from the AV node, through the fibrous skeleton, into the interventricular septum.**
3. **The AV bundle divides into right and left bundle branches, and action potentials descend to the apex of each ventricle along the bundle branches.**
4. **Action potentials are carried by the Purkinje fibres from the bundle branches to the ventricular walls.**

Structure	Location	Function
Sinoatrial Node (SA Node)	Wall of the right atrium near superior vena cava	Acts as the pacemaker; initiates electrical impulses causing atrial contraction
Atrioventricular Node (AV Node)	Interatrial septum near tricuspid valve	Receives impulses from SA node and delays transmission to allow complete atrial contraction
Atrioventricular Bundle (Bundle of His)	Upper part of interventricular septum	Transmits impulses from AV node to the apex of the heart via bundle branches
Purkinje Fibres (Conducting Myofibrils)	Inner walls of ventricles (subendocardial layer)	Distribute impulses rapidly through ventricular walls causing ventricular contraction

SLOs Based (MCQs)

PHASES OF CARDIAC CYCLE AND CONDCUTION SYSTEM OF HEART

1. Which of the following events occurs during ventricular systole?
A. The atria contract
B. Atrioventricular valves open
C. Semilunar valves close
D. Semilunar valves open

2. Which phase of the cardiac cycle allows atrial blood to fill the ventricles?
A. Ventricular systole B. Atrial diastole
C. Atrial systole D. Ventricular diastole

3. What causes the atrioventricular valves to close during the cardiac cycle?
A. Atrial contraction
B. Ventricular relaxation
C. High pressure in the ventricles
D. Opening of semilunar valves

4. Which part of the cardiac conduction system delays the impulse to allow ventricular filling?
A. SA node B. AV node
C. Bundle of His D. Purkinje fibres

5. Why is the SA node referred to as the pacemaker of the heart?
A. It has the thickest walls
B. It is located in the left atrium
C. It generates impulses at the highest frequency
D. It delays impulses to the ventricles

6. In which phase are the semilunar valves closed and atrioventricular valves open?
A. Atrial systole
B. Ventricular systole
C. Ventricular diastole
D. Isovolumetric contraction

7. Which structure conducts impulses most rapidly in the heart?
A. SA node B. AV node
C. Bundle branches D. Purkinje fibres

8. Which of the following correctly describes the delay of electrical conduction at the AV node?
A. 0.04 seconds B. 0.11 seconds
C. 0.01 seconds D. 0.20 seconds

Answer Key

1. D	2. C	3. C	4. B	5. C	6. C	7. D	8. B

Short Questions

Q.1 Explain why the atrioventricular (AV) nodal delay is essential in the cardiac cycle.

Ans. The AV nodal delay is a brief pause of approximately 0.11 seconds in the conduction of electrical impulses from the atria to the ventricles. This delay is vital because it allows the atria to complete their contraction before the ventricles begin contracting.

As a result, the ventricles receive the maximum possible volume of blood from the atria, optimizing stroke volume and ensuring efficient cardiac output. Without this delay, atrial and ventricular contractions could overlap, leading to incomplete ventricular filling and reduced efficiency of blood circulation. This coordinated timing between atrial systole and ventricular systole is a key feature of a well-functioning cardiac cycle.

> **MARKING SCHEME (3 Marks)**
> - Duration of delay (0.11 seconds) — 1 mark
> - Reason: ensures atrial contraction completes before ventricular systole — 2 marks

Q.2 Describe the role and location of Purkinje fibres in the heart's conduction system.

Ans. Purkinje fibres are specialized, large-diameter cardiac muscle fibres located in the subendocardial layer of the ventricular walls. They form the terminal branches of the heart's conduction system, arising from the right and left bundle branches. Their primary function is to rapidly conduct action potentials to the ventricular myocardium, ensuring that the depolarization signal reaches all parts of the ventricles almost simultaneously. This rapid conduction allows both ventricles to contract together, producing a strong and synchronized pumping action that efficiently propels blood into the pulmonary artery and aorta. The structural specialization of Purkinje fibres — including abundant gap junctions and fewer myofibrils — enhances their ability to transmit impulses quickly and reliably.

> **MARKING SCHEME (3 Marks)**
> - Description of Purkinje fibres — 1 mark
> - Location — 1 mark
> - Function — 1 mark

Q.3 Outline the sequence of electrical impulse conduction through the heart.

Ans. The cardiac conduction system begins with the **sinoatrial (SA) node** in the right atrium, which generates an electrical impulse that spreads across both atria, causing atrial contraction. The impulse then reaches the **atrioventricular (AV) node**, where it experiences a brief AV nodal delay. From the AV node, the signal travels through the **atrioventricular bundle (Bundle of His)**, which passes into the interventricular septum. Here, it divides into the **right** and **left bundle branches**, which conduct impulses toward the apex of the heart. Finally, the signal spreads through the **Purkinje fibres**, rapidly depolarizing the ventricular muscle and triggering a coordinated ventricular contraction. This precise sequence ensures that atrial systole is followed by ventricular systole, optimizing blood flow.

> **MARKING SCHEME (3 Marks)**
> - Correct names of all conduction components — *1 mark*
> - Correct sequence of conduction - 2 marks

Q.4 Explain the function of heart valves during ventricular systole.

Ans. During ventricular systole, rising ventricular pressure forces the **atrioventricular (AV) valves** — the bicuspid (mitral) and tricuspid valves — to close. This prevents the backflow of blood into the atria. At the same time, the increasing pressure exceeds that in the aorta and pulmonary artery, causing the **semilunar valves** — the aortic and pulmonary valves — to open. This allows blood to be ejected from the right ventricle into the pulmonary artery and from the left ventricle into the aorta. The precise timing of these valve actions ensures one-way blood flow and maintains high efficiency in the cardiac cycle. Once ventricular pressure falls below arterial pressure, the semilunar valves close to prevent backflow, marking the end of systole.

> **MARKING SCHEME (3 Marks)**
> - Function of atrioventricular valves — 1.5 marks
> - Function of aortic and pulmonary valves — 1.5 marks

Q.5 How does atrial systole contribute to the cardiac cycle?

Ans. Atrial systole is the contraction of the atria, which pushes the remaining blood into the ventricles after passive filling. This ensures the ventricles are fully loaded before ventricular systole begins, enabling efficient blood ejection during contraction.

> **MARKING SCHEME (3 Marks)**
> - Contribution of atrial contraction – 1 mark
> - Contribution of ventricular relaxation – 1 mark

Q.6 What structural feature of the SA node makes it the natural pacemaker of the heart?

Ans. The SA node contains specialized cardiac muscle fibres capable of *spontaneously generating action potentials*. It has the *highest rate of* impulse generation compared to other conduction system components, making it the primary initiator of the heartbeat under normal conditions.

> **MARKING SCHEME (3 Marks)**
> - SA node contains specialized cardiac muscle fibres – 1 mark
> - Highest action potential production – 2 marks

Q.7 Explain what occurs during ventricular diastole.

Ans. During ventricular diastole, the ventricles relax, causing intraventricular pressure to drop. This leads to the closure of semilunar valves to prevent backflow from the arteries and the opening of atrioventricular valves to allow blood to flow from the atria into the ventricles.

MARKING SCHEME (3 Marks)
- Function of aortic/pulmonary semilunar valves – 1.5 marks
- Function of atrioventricular valves – 1.5 marks

Q.8 What is the importance of an artificial pacemaker, where is it placed in the heart, and how long does it function accurately?

Ans. An artificial pacemaker regulates abnormal heart rhythms, particularly in bradycardia. It is implanted under the skin near the left collarbone and connected to heart chambers via insulated leads. The device sends electrical impulses to maintain normal rhythm and typically works accurately for 5–15 years.

MARKING SCHEME (3 Marks)
- Importance – 1 mark
- Placement – 1 mark
- Duration – 1 mark

SLO [B-12-R-41]
List the principles and uses of Electrocardiogram.

ELECTROCARDIOGRAM

Principles

An **Electrocardiogram (ECG)** is a diagnostic tool that graphically records the **electrical activity of the heart** over time. It is based on the fact that every heartbeat is initiated and coordinated by **electrical impulses** generated within specialized cardiac muscle tissue.

1. **Generation of Electrical Impulses**
   - During each cardiac cycle, the **sinoatrial (SA) node** generates an impulse that spreads through the atria, causing **atrial depolarization.**
   - The impulse then travels to the **atrioventricular (AV) node,** bundle of His, and Purkinje fibers, stimulating **ventricular depolarization and contraction,** followed by **repolarization** to restore resting potential.
2. **Transmission through Body Tissues**
   - These electrical changes spread not only through the heart muscle but also through the surrounding body fluids and tissues, eventually reaching the skin surface.
3. **Detection by Electrodes**
   - **Electrodes** placed on specific standardized points of the skin (limbs and chest) detect the tiny voltage changes produced by the heart.
   - Multiple electrodes together form **leads**, which record electrical activity from different angles, giving a complete picture of heart function.
4. **Amplification and Recording**
   - The small electrical signals are **amplified** by the ECG machine.
   - They are then displayed as a **graphical tracing** on paper or a monitor, where different waves and intervals correspond to specific phases of the heartbeat.

▶ **Main ECG Waves**
- **P wave** → Represents **atrial depolarization**, i.e., electrical activity leading to atrial contraction.
- **QRS complex** → Represents **ventricular depolarization**, which triggers ventricular contraction. (Atrial repolarization also occurs here but is masked by the stronger ventricular signal.)
- **T wave** → Represents **ventricular repolarization**, where the ventricles recover and prepare for the next cycle.

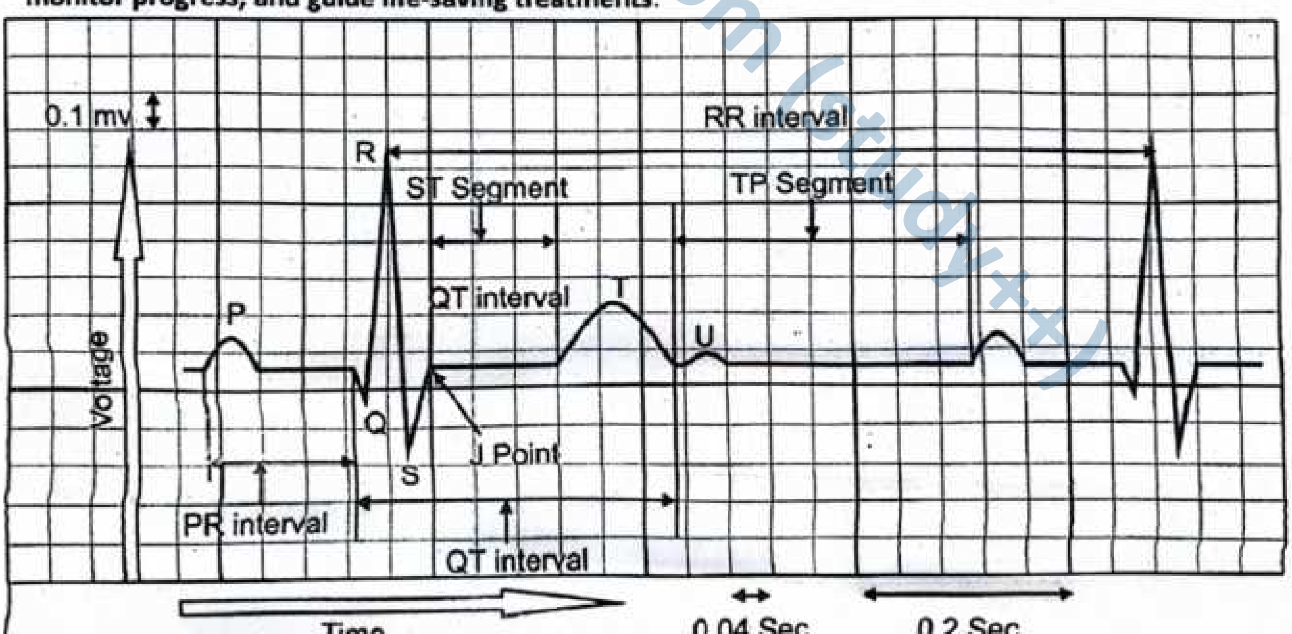
Depolarization is the process in which cardiac muscle cells lose their resting negative charge, triggering contraction.
Repolarization is the restoration of the resting negative charge in these cells, leading to muscle relaxation.
In an ECG, **depolarization** generates the **P wave** (atria) and **QRS complex** (ventricles), while **repolarization** produces the **T wave.**

Uses

The **Electrocardiogram (ECG)** is one of the most widely used, quick, and non-invasive diagnostic tests in medicine. It provides valuable information about the **electrical activity, rhythm, and overall function of the heart**, making it essential in both clinical and emergency settings.

1. **Diagnosis of Heart Disorders**
   - **Arrhythmias** → ECG identifies irregular heart rhythms, such as tachycardia (fast rate), bradycardia (slow rate), and fibrillation (uncoordinated beating).
   - **Conduction Abnormalities** → Detects heart block, where electrical signals are delayed or blocked in conduction pathways.
   - **Myocardial Infarction (Heart Attack)** → Characteristic changes in the ST segment and T wave indicate areas of dead or damaged heart tissue.
   - **Ischemia** → Reduced blood flow to the heart muscle can be revealed by subtle wave pattern changes.
2. **Monitoring Heart Health**
   - Used during **routine medical check-ups** for patients with risk factors like hypertension, diabetes, smoking, or family history of heart disease.
   - Assesses cardiac performance during **exercise stress tests**, where ECG is recorded while the patient runs or cycles.
   - Monitors **post-surgical recovery** to ensure the heart is regaining normal rhythm and function.
3. **Guiding Treatment**
   - Provides information to guide decisions regarding **pacemaker insertion**, choice of **anti-arrhythmic or cardiac drugs**, or the need for **surgical interventions**.
   - Tracks the **effectiveness of ongoing treatment**, such as medication for arrhythmias, anticoagulants, or therapy after a heart attack.
4. **Emergency Medicine**
   - In hospital emergency departments or ambulances, ECG is used for the **rapid evaluation of chest pain** patients.
   - Quickly detects **life-threatening conditions** (e.g., ventricular fibrillation, massive myocardial infarction) that require immediate intervention such as defibrillation or urgent surgery.

The ECG is a **non-invasive, quick, and reliable** method to assess the heart's electrical activity. By translating invisible electrical signals into visible waveforms, it allows healthcare professionals to **detect problems early, monitor progress, and guide life-saving treatments**.



• PR interval	0.12-0.20 sec	• QT interval	0.4-0.43 sec
• QRS duration	0.08-0.10 sec	• RR interval	0.6-1.0 sec

Figure: Electrocardiogram (ECG)

SLOs Based (MCQs)

ECG

5

1. **Which wave of the ECG corresponds to ventricular depolarization?**
   A. P wave B. T wave
   C. QRS complex D. U wave

2. **What does the ST segment in an ECG represent?**
   A. Atrial depolarization
   B. Ventricular repolarization
   C. Time between ventricular depolarization and repolarization
   D. Delay in SA node conduction

3. **An elevated ST segment in an ECG is typically a sign of:**
   A. Heart murmur B. Myocardial infarction
   C. Ventricular fibrillation D. Hypotension

4. **What is recorded during the P wave of the ECG?**
   A. Ventricular depolarization B. Atrial depolarization
   C. Ventricular repolarization D. Atrial repolarization

5. **Which instrument is used to record the ECG?**
   A. Sphygmomanometer B. Stethoscope
   C. Electrocardiograph D. Cardiometer

Answer Key

1. C	2. C	3. B	4. B	5. C

Short Questions

Q.1 What is an electrocardiogram (ECG)?

Ans. An electrocardiogram (ECG) is the graphical recording of the electrical activity of the heart during the cardiac cycle. It is obtained using an electrocardiograph through electrodes placed on the skin. It records depolarization and repolarization of cardiac muscle fibres to assess heart activity.

MARKING SCHEME (3 Marks)
- Definition – 1 mark
- Function – 2 marks

Q.2 What do the waves in an ECG represent?

Ans.
- **P wave:** Represents atrial depolarization, which precedes atrial contraction.
- **QRS complex:** Represents ventricular depolarization, which precedes ventricular contraction.
- **T wave:** Represents ventricular repolarization, occurring just before ventricular relaxation.

MARKING SCHEME (3 Marks)
- P wave – 1 mark
- QRS complex – 1 mark
- T wave – 1 mark

Q.3 What is the clinical importance of an ECG?

Ans. An ECG is a vital, non-invasive diagnostic tool to evaluate heart function. It helps in detecting arrhythmias, myocardial infarction, and conduction blocks. It also identifies chamber enlargement, electrolyte imbalances, and monitors artificial pacemaker activity or the effects of cardiac drugs.

MARKING SCHEME (3 Marks)
- Any three correct applications – 1 mark each

Q.4 What is the significance of the ST segment and T wave in an ECG?

Ans. The ST segment represents the interval between ventricular depolarization and repolarization. Its elevation or depression is a key indicator of myocardial ischemia or injury. The T wave shows ventricular repolarization, and abnormal patterns may suggest electrolyte imbalance or cardiac disease.

MARKING SCHEME (3 Marks)
- Significance of ST segment – 1.5 marks
- Significance of T wave – 1.5 marks

Q.5 What causes the QRS complex in an ECG and what is its timing relevance?

Ans. The QRS complex is caused by ventricular depolarization. It is composed of a small downward Q wave, a tall upward R wave, and a downward S wave. It corresponds to ventricular contraction and, in a healthy heart, appears narrow and rapid, reflecting efficient conduction.

MARKING SCHEME (3 Marks)
- Cause of QRS complex – 1.5 marks
- What it represents – 1.5 marks

SLO [B-12-R-24]
Describe the detailed structure of arteries, veins and capillaries.

BLOOD VESESLS

The circulatory system comprises three main types of blood vessels: **arteries (and arterioles)**, which carry blood *away from the heart*; **veins**, which return blood *to the heart*; and **capillaries**, which allow exchange of materials between blood and tissues.

Arteries and Arterioles

Arteries are muscular, elastic vessels that transport blood away from the heart. They appear pink and are embedded deep within muscles. The largest artery, the **aorta**, measures approximately **23 mm** in diameter, while **arterioles** are smaller branches (~**0.2 mm**), delivering blood to capillaries.

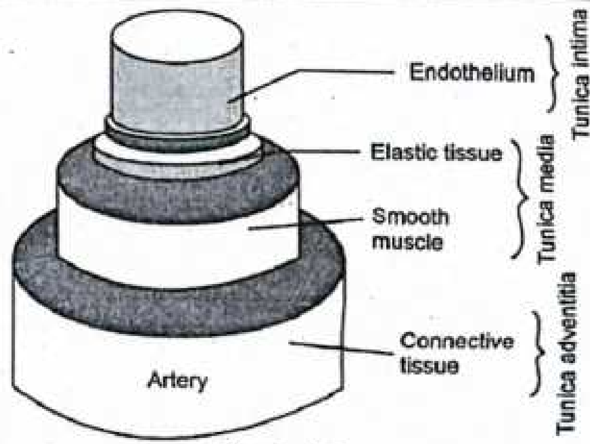


Figure: Artery

▶ Structure of Arteries

The arterial wall consists of three concentric layers:

- **Tunica intima:** Inner layer made of simple squamous epithelium and elastin-rich elastic fibres.
- **Tunica media:** Middle muscular layer composed of circular smooth muscle and elastic fibres.
- **Tunica adventitia:** Outer layer of white fibrous connective tissue providing structural support.

Unlike veins, arteries **lack valves** due to high pressure and unidirectional blood flow.

Capillaries

Capillaries are the thinnest blood vessels, enabling efficient exchange of gases, nutrients, and waste with tissues. Their walls are composed of a **single layer of endothelial cells,** allowing rapid diffusion. Most capillaries are **7–9 µm** in diameter and about **1 mm** long. Red blood cells pass through in **single file**, ensuring maximal surface contact.

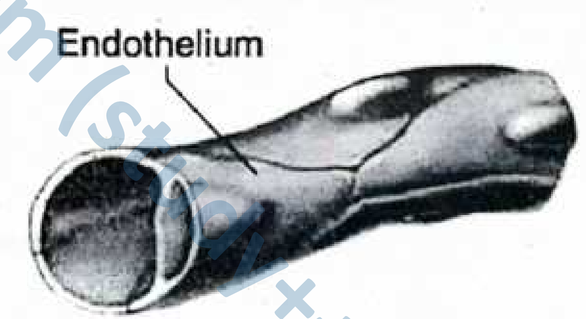


Figure: Capillary

Veins and Venules

Veins carry blood back to the heart under low pressure. They are located more superficially than arteries and appear **bluish** under the skin. Like arteries, veins have three layers:

- **Tunica intima:** Inner endothelial lining with thin elastic fibres.
- **Tunica media:** Thin layer of smooth muscle, collagen, and few elastic fibres.
- **Tunica adventitia:** Collagen-rich connective tissue forming the outer layer.

Venules (40–50 µm in diameter) collect blood from capillaries and lead into small veins.

▶ Valves in Veins

Veins with diameters greater than **2 mm** contain **valves** that ensure unidirectional blood flow **toward the heart.** These valves are mainly present **below the heart**, especially in the abdomen and lower limbs. Above the heart, gravity aids blood return, so **valves are absent**. Muscular contraction and **semilunar valves** support venous return under low pressure.

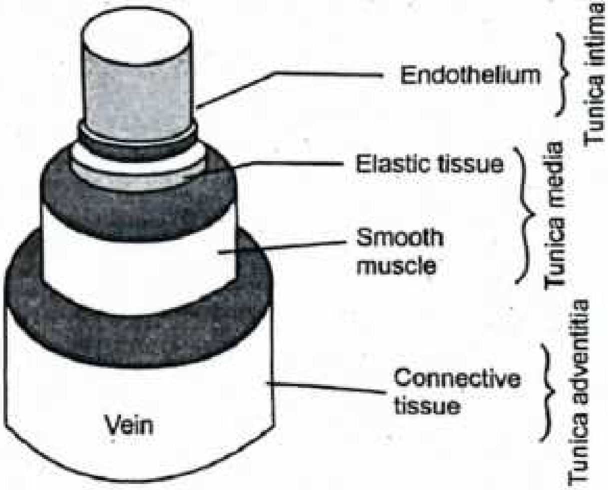


Figure: Vein

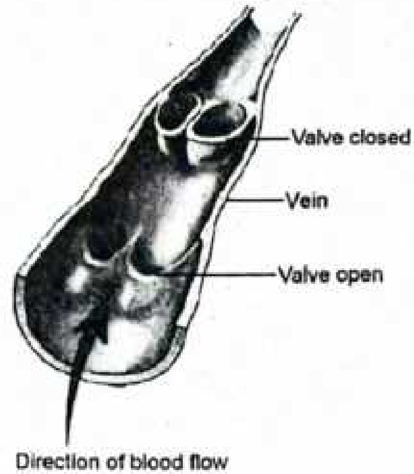


Figure: Valves in veins

Feature	Arteries	Veins	Capillaries
Function	Carry blood away from the heart	Carry blood toward the heart	Exchange of gases, nutrients, and waste with tissues
Type of Blood	Mostly oxygenated (except pulmonary artery)	Mostly deoxygenated (except pulmonary vein)	Both; transition between oxygenated and deoxygenated
Wall Thickness	Thick, muscular, and elastic	Thinner, less muscular	Extremely thin (one-cell thick)
Lumen Size	Narrow	Wide	Very narrow (just wide enough for RBCs)
Valves	Absent (except in pulmonary artery)	Present to prevent backflow	Absent
Blood Pressure	High	Low	Drops progressively through capillary beds
Location	Deep within body	Often near body surface	Found within tissues, forming extensive networks

SLO [B-12-R-43]
Describe the role of arterioles in vasoconstriction and vasodilation.

ROLE OF ARTERIOLES IN VASO-CONSTRICTION AND VASODILATION

Arterioles are small branches of arteries that directly lead into capillaries. They are often called **"resistance vessels"** because they regulate blood flow and blood pressure by adjusting their diameter. The muscular walls of arterioles, composed mainly of **smooth muscle**, make them key regulators of circulation.

1. Vasoconstriction

Definition

- *Vasoconstriction* refers to the **narrowing of the lumen (internal space)** of arterioles. This occurs when the smooth muscle fibers in the arteriole walls contract. Because arterioles are the main regulators of blood flow, even a small degree of narrowing has a **major impact on blood circulation and blood pressure.**

Mechanism

- Vasoconstriction is primarily under the control of the **sympathetic nervous system**. When the body is under stress, fear, *or cold, nerve* endings release **norepinephrine**, which binds to receptors on smooth muscle *cells*, causing contraction.

- Certain **hormones** like **angiotensin II, vasopressin, and epinephrine** also promote vasoconstriction to raise blood pressure during fluid loss or shock.
- **Local chemical signals** (e.g., reduced oxygen or changes in pH) can also influence vasoconstriction in specific tissues.

Effects

- **Reduced blood flow:** Less blood reaches the tissue supplied by the constricted arteriole.
- **Increased vascular resistance:** Narrower vessels make it harder for blood to flow, which raises **systemic blood pressure**.
- **Blood redistribution:** Blood is diverted away from less active organs (e.g., skin, digestive tract) toward vital organs (e.g., heart, brain).

Example

- During **cold weather**, skin arterioles undergo vasoconstriction. This reduces warm blood flow to the skin surface, minimizing heat loss and helping the body maintain its core temperature.

2. Vasodilation

Definition

- Vasodilation is the **widening of the lumen** of arterioles, caused by relaxation of their smooth muscle layer. This process increases the capacity of blood vessels and reduces resistance to blood flow, making it vital for supplying active tissues with nutrients and oxygen.

Mechanism

- Vasodilation is stimulated by the **parasympathetic nervous system** and chemical signals from tissues that are metabolically active.
- **Local metabolites** such as **CO_2, lactic acid, and adenosine** act as vasodilators, signaling that tissues require more oxygen.
- **Vasodilator chemicals** like **nitric oxide (NO)**, released by endothelial cells, and **histamine**, released during inflammation or allergic reactions, relax smooth muscles and widen arterioles.

Effects

- **Increased blood flow:** More oxygen, glucose, and nutrients are delivered to tissues.
- **Decreased vascular resistance:** Wider vessels reduce systemic blood pressure, aiding in homeostatic regulation.
- ***Enhanced waste removal:*** Increased flow helps remove CO_2 and metabolic by-products more efficiently.

Example

- During **exercise**, arterioles in skeletal muscles dilate. This increases blood supply to meet the higher oxygen and nutrient demands of active muscle fibers. At the same time, vasoconstriction may occur in less active areas (e.g., digestive organs), ensuring blood is prioritized for muscles.

The ability of arterioles to undergo **vasoconstriction and vasodilation** makes them crucial regulators of **blood flow distribution, systemic blood pressure, and body temperature**. They act like "control valves" that open or close according to the body's immediate needs, ensuring efficient circulation and homeostasis.

3. Functional Importance of Arterioles

Arterioles are often called the **"resistance vessels"** of the circulatory system because their small size and muscular walls give them the unique ability to control blood distribution and regulate blood pressure. Their role extends far beyond simply carrying blood — they are **dynamic regulators of circulation and homeostasis.**

1. Control of Blood Distribution

- Arterioles act as **control valves** that determine how much blood reaches each organ or tissue.
- When a tissue becomes metabolically active (e.g., muscles during exercise), its arterioles dilate to increase blood flow. Conversely, in resting or less active tissues, arterioles constrict, reducing blood supply.
- This selective distribution ensures that **oxygen and nutrients** are delivered according to demand *while conserving energy.*

2. Regulation of Systemic Blood Pressure

- By adjusting their diameter, arterioles directly influence **peripheral resistance** (the resistance blood faces as it flows through vessels).

- Widespread vasoconstriction raises systemic blood pressure, while widespread vasodilation lowers it.
- In this way, arterioles help maintain a **stable blood pressure**, which is essential for effective perfusion of vital organs such as the brain, heart, and kidneys.

3. Role in Thermoregulation

- Arterioles in the skin play a key part in regulating **body temperature.**
- During heat stress, they undergo vasodilation, allowing more warm blood to reach the skin surface, where heat is lost through radiation and sweating.
- In cold conditions, skin arterioles constrict to reduce blood flow, conserving heat for core organs.
- This adjustment maintains the body's **thermal balance** in changing environments.

Q. Justify how Vasodilation and Vasoconstriction is Reflective of Emotions?

Ans. During emotional rage such as apprehension and rage vasodilation occurs due to secretion of epinephrine. It is a hormone that is responsible for fear, flight and fright conditions. The sympathetic vasodilator fibres are part of a regulatory system that originates in cerebral cortex and ends at postganglionic neurons in blood vessels on skeletal muscles, activate them to release acetylcholine, and vasodilation occurs. Blood discharge through thoroughfare channels rather than capillaries so heat loss occurs and the skin becomes hot and red. While in vasoconstriction blood supply becomes less to skin, so heat is preserved and the skin becomes cold. Situations such as shock, hypotension and tachycardia occur by stimulation of arterial stretch receptors and production of hypertension and bradycardia (slowness of the heart) occur by increased intracranial pressure.

SLO [B-12-R-44]
Describe the role of pre-capillary sphincters in regulating the flow of blood through the capillaries.

ROLE OF PRE-CAPILLARY SPHINCTERS

Pre-capillary sphincters are **rings of smooth muscle** located at the point where arterioles branch into capillaries. They act as **tiny valves** that regulate the entry of blood into the capillary networks of tissues.

1. Structure and Location

Position in the Microcirculation:

Pre-capillary sphincters are located at the point where **metarterioles** (small vessels branching from arterioles) give rise to capillaries. This makes them the **gateways** that determine whether blood enters a particular capillary bed.

Muscular Composition:

They are made up of **circular rings of smooth muscle fibers.** Unlike skeletal muscles, smooth muscles contract involuntarily, meaning their activity is not under conscious control. These muscle fibers tightly encircle the capillary openings, enabling them to act like adjustable valves.

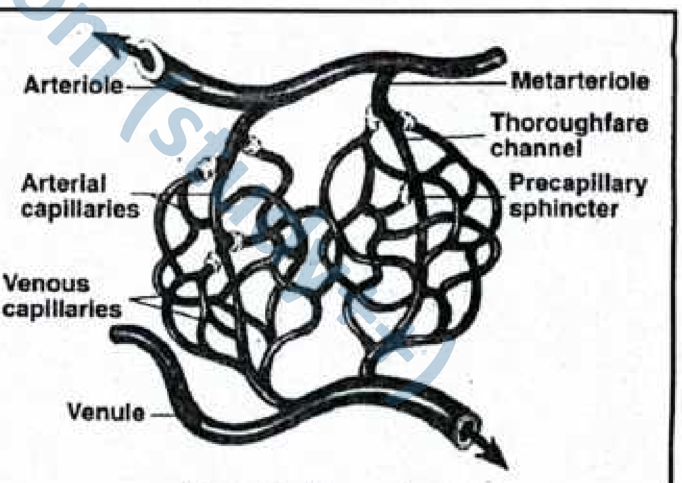


Figure: Capillary network

Microscopic Size:

Pre-capillary sphincters are extremely small (microscopic structures), yet their effect on blood flow is profound because they determine the perfusion of entire capillary networks.

Regulatory Control:

Their activity is influenced by:

- **Local Chemical Signals:** Accumulation of carbon dioxide, lactic acid, hydrogen ions, or other metabolites in active tissues stimulates relaxation, opening the sphincters.

- **Oxygen Levels:** Low oxygen concentration in tissues triggers opening to allow more blood supply.
- **Nervous Input:** The autonomic nervous system, especially sympathetic fibers, can cause contraction or relaxation of sphincters depending on the body's needs (e.g., stress vs. rest).
- **Hormones and Mediators:** Substances like adrenaline, nitric oxide, and histamine can also modulate sphincter activity.

2. Function in Blood Flow Regulation

Contraction (Closure of Sphincters):

- When pre-capillary sphincters contract, they **narrow or completely close the entrance** to capillary beds. This diverts blood away from inactive tissues, sending it instead through the **thoroughfare channel** (a direct vascular pathway connecting the arteriole to the venule).
    - **Purpose:** Prevents unnecessary energy expenditure and avoids excessive blood perfusion in tissues that are temporarily at rest, such as skeletal muscles during sleep or digestion when physical activity is low.
    - **Physiological Example:** During cold conditions, sphincters in the skin capillaries constrict to reduce heat loss by limiting blood flow to the skin surface.

Relaxation (Opening of Sphincters):

- When sphincters relax, they **open the gateway to the capillary network**, allowing a rich supply of blood into the surrounding tissue. This increases the **exchange of gases, nutrients, and wastes** between the blood and tissue cells.
    - **Triggers:** Relaxation occurs when oxygen levels are low, carbon dioxide concentration is high, or when metabolic by-products like lactic acid accumulate.
    - **Purpose:** Ensures that metabolically active tissues — such as exercising muscles or actively secreting glands — receive the oxygen and nutrients they urgently require.
    - **Physiological Example:** During exercise, precapillary sphincters in muscle tissues relax widely, allowing increased blood flow to meet the high demand for oxygen and glucose.

Pre-capillary sphincters act as **local regulators of microcirculation**, continuously adjusting **blood distribution** according to tissue needs. They ensure that the body maintains a delicate balance — conserving blood in resting tissues while prioritizing supply to active or stressed regions.

3. Functional Importance

Tissue Perfusion Control:

- Pre-capillary sphincters act as "gatekeepers" of the microcirculation. By opening or closing, they ensure that **blood is delivered primarily to tissues with high metabolic activity**. For example, active muscles or secretory glands receive increased blood supply, while less active tissues receive minimal flow.

Efficient Blood Distribution:

- Since the total volume of blood in the body is limited, not all capillary beds can remain open at the same time. These sphincters prevent unnecessary perfusion of resting tissues and **redirect blood toward areas where it is urgently required**, ensuring efficient use of cardiac output.

Homeostasis Maintenance:

- By regulating the entry of blood into capillaries, sphincters play a crucial role in **maintaining stable internal conditions**. They help balance oxygen delivery, nutrient availability, and removal of carbon dioxide and metabolic wastes, thereby preserving the chemical environment needed for normal cell function.

Adaptive Response:

- **During Exercise:** Pre-capillary sphincters in skeletal muscles relax widely, allowing a surge of blood to meet the increased **oxygen and glucose demand** of contracting muscle fibers.
- **During Rest:** Many sphincters remain constricted, conserving blood flow for vital organs such as the heart, brain, and kidneys, which must remain active continuously.

Pre-capillary sphincters are crucial gatekeepers that **open or close capillary beds** depending on tissue needs. By doing so, they ensure **efficient distribution of blood**, maintain **homeostasis**, and allow the body to adapt quickly to changes in activity or environment.

SLOs Based (MCQs)

TYPES OF BLOOD VESSELS, ROLE OF ARTERIOLES IN VASO-CONSTRICTION AND VASODILATION AND ROLE OF PRE-CAPILLARY SPHINCTERS

1. **Which structural feature is *absent* in arteries but *present* in veins?**
   A. Tunica intima B. Elastic fibres
   C. Smooth muscle D. Valves
2. **Which blood vessel permits exchange of materials between blood and tissues?**
   A. Arteries B. Veins
   C. Capillaries D. Arterioles
3. **Vasodilation resul in:**
   A. Decrease in vessel diameter
   B. Increase in blood pressure
   C. Increase in vessel diameter
   D. Closure of capillaries
4. **Which hormone is a potent vasoconstrictor?**
   A. Histamine B. Norepinephrine
   C. Prostaglandin E2 D. Serotonin
5. **Precapillary sphincters regulate blood flow:**
   A. Between veins and venules
   B. Between arteries and arterioles
   C. Into individual capillaries
   D. Out of the heart chambers

Answer Key

1. D	2. C	3. C	4. B	5. C

Short Questions

Q.1 What are the three main types of blood vessels in the human circulatory system?

Ans. The three main types of blood vessels are:
- **Arteries and arterioles:** Carry blood away from the heart.
- **Veins and venules:** Return blood to the heart.
- **Capillaries:** Allow exchange of gases, nutrients, and wastes between blood and tissues.

MARKING SCHEME (3 Marks)
- Arteries/arterioles – 1 mark
- Veins/venules – 1 mark
- Capillaries – 1 mark

Q.2 Describe the key structural difference between arteries and veins.

Ans. Arteries have thick muscular walls, a narrow lumen, and no valves, while veins have thin walls, a wide lumen, and contain valves. Functionally, arteries carry blood under high pressure away from the heart, while veins return blood under low pressure and prevent backflow with valves.

MARKING SCHEME (3 Marks)
- Structural difference – 1 mark
- Functional difference – 2 marks

Q.3 What is vasoconstriction, and how does it affect blood flow?

Ans. Vasoconstriction is the narrowing of arterioles due to smooth muscle contraction. It decreases blood flow to tissues and raises blood pressure. This response may be triggered by hormones such as norepinephrine.

MARKING SCHEME (3 Marks)
- Definition – 1 mark
- Function/effect – 2 marks

Q.4 Explain vasodilation and name one vasodilator agent.

Ans. Vasodilation is the widening of blood vessels due to relaxation of smooth muscle in arterioles. It increases blood flow and oxygen supply to active tissues. Histamine is one example of a vasodilator agent.

MARKING SCHEME (3 Marks)
- Definition – 2 marks
- Example of vasodilator – 1 mark

Q.5 What are precapillary sphincters, and what is their role in circulation?

Ans. Precapillary sphincters are circular smooth muscles located at the openings of capillaries. They regulate blood flow into capillaries by contracting or relaxing according to tissue demand. Their rhythmic action, called vasomotion, ensures efficient blood distribution.

MARKING SCHEME (3 Marks)
- Structure – 1 mark
- Location – 1 mark
- Function – 1 mark

Q.6 Why are valves present in veins but not in arteries?

Ans. Valves are present in veins, especially in the limbs, to prevent backflow and aid blood return to the heart under low pressure. Arteries do not require valves because blood flows at high pressure and in one direction from the heart.

MARKING SCHEME (3 Marks)
- Function in veins – 2 marks
- Function in arteries – 1 mark

SLO [B-12-R-45]
Trace the path of the blood through the pulmonary and systemic circulation (coronary, hepatic-portal and renal circulation).

PATH OF BLOOD THROUGH THE PULMONARY AND SYSTEMIC CIRCULATION

The circulatory system operates as a **double circulation**: blood passes through the heart **twice in one complete cycle**—once through the lungs (pulmonary circulation) and once through the body (systemic circulation). This ensures continuous oxygenation and delivery of nutrients to tissues, as well as the removal of wastes.

1. Pulmonary Circulation

Pulmonary circulation is the pathway that carries **deoxygenated blood from the heart to the lungs** for gas exchange and then returns **oxygenated blood back to the heart.** It ensures that blood is continuously recharged with oxygen before it is pumped to the rest of the body.

▶ Pathway

1. The circulation begins in the **right ventricle**, which contracts and pumps **deoxygenated blood** into the **pulmonary trunk**.
2. The pulmonary trunk quickly divides into the **right and left pulmonary arteries**, each carrying blood to the respective lung.
3. Within the lungs, these arteries branch into **smaller arterioles and capillaries** that surround the alveoli (tiny air sacs).
4. At the alveolar-capillary interface, **gas exchange** occurs:
   - o **Carbon dioxide** diffuses out of the blood into the alveoli to be exhaled.
   - o **Oxygen** diffuses from the alveoli into the blood.
5. Now oxygenated, the blood is collected by small venules that join to form the **pulmonary veins.**
6. A total of **four pulmonary veins** (two from each lung) carry this oxygen-rich blood back to the heart.
7. These veins empty into the **left atrium**, completing the pulmonary circuit.

▶ Key Function

Pulmonary circulation **restores oxygen and removes carbon dioxide**, preparing blood for **systemic circulation**, where oxygen and nutrients are delivered to body tissues.

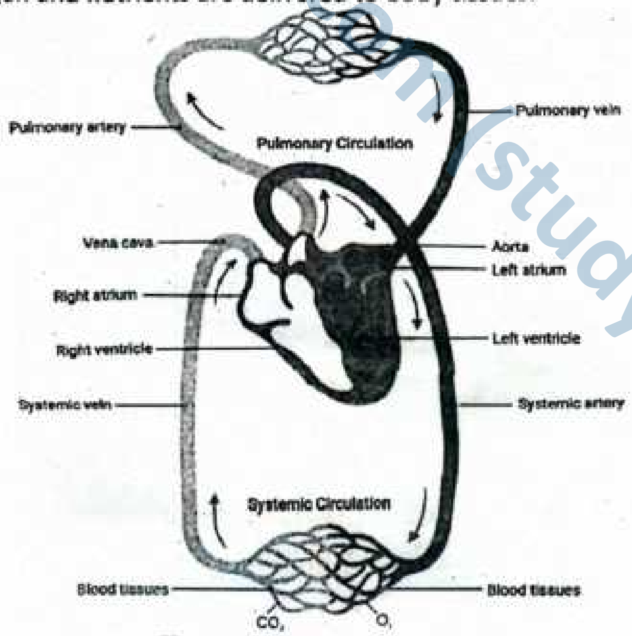


Figure: Cardiovascular system

2. Systemic Circulation

Systemic circulation is the pathway that delivers **oxygenated blood from the heart to the entire body** and returns **deoxygenated blood back to the heart**. It operates under high pressure and ensures that every tissue receives the oxygen and nutrients it needs for metabolism.

► Pathway

1. The circulation begins in the **left ventricle**, whose strong muscular wall pumps oxygen-rich blood into the **aorta** with great force.
2. The **aorta** branches into major systemic arteries that supply the head, arms, trunk, and lower limbs. These arteries further divide into **smaller arterioles and capillaries** in all body tissues.
3. In the **systemic capillaries**, exchange takes place:
   - o **Oxygen and nutrients** (e.g., glucose, amino acids, hormones) diffuse from the blood into the tissues.
   - o **Carbon dioxide, urea, and other wastes** move from the tissues into the blood.
4. The now deoxygenated blood is collected by venules, which merge into **systemic veins**.
5. Veins from the upper body drain into the **superior vena cava**, while those from the abdomen, pelvis, and lower limbs drain into the **inferior vena cava**.
6. Both venae cavae deliver blood into the **right atrium**, completing the systemic circuit.

► Key Function

Systemic circulation ensures the **supply of oxygen, nutrients, and hormones** to all body cells, while also **collecting carbon dioxide and metabolic wastes** for removal through the lungs, kidneys, and other excretory organs.

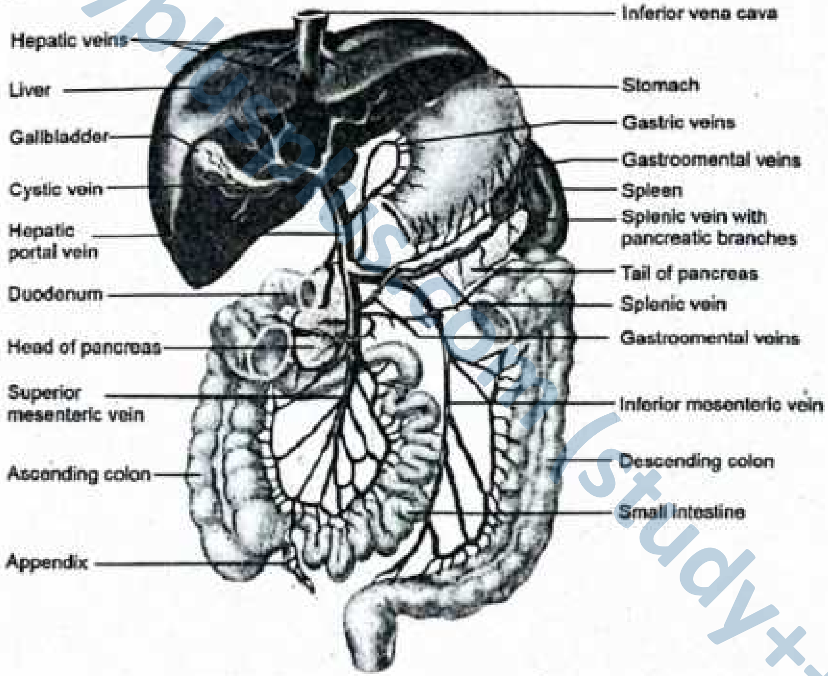


Figure: Hepatic portal system

3. Specialized Circulations within the Systemic Pathway

a) Coronary Circulation (Blood supply of the heart itself)

The heart is a muscular organ that works tirelessly, beating about 70–80 times per minute throughout life. Because of this continuous activity, the **myocardium (cardiac muscle)** requires a constant and rich blood supply. This is provided by the **coronary circulation**, a specialized branch of systemic circulation dedicated *to the heart* itself.

► Pathway

1. Arterial Supply

- o Oxygenated blood leaves the **ascending aorta** almost immediately after it emerges from the left ventricle.

- Two main vessels, the **right coronary artery** and the **left coronary artery**, branch off the aorta and spread around the heart.
- These arteries give rise to smaller branches (e.g., circumflex artery, interventricular branches) that penetrate deep into the myocardium to nourish every muscle fiber.

2. **Venous Return**
   - After gas and nutrient exchange, **deoxygenated blood** from the myocardium is collected by **cardiac veins.**
   - These veins converge into a large channel called the **coronary sinus**, located on the posterior side of the heart.
   - The coronary sinus then empties directly into the **right atrium**, where deoxygenated blood from the entire body also arrives.

▶ **Key Function**

Coronary circulation ensures that the **heart muscle receives continuous oxygen and nutrient supply** while simultaneously removing carbon dioxide and wastes. Any blockage in this system (e.g., coronary artery disease) can lead to **angina (chest pain)** or even a **myocardial infarction (heart attack),** highlighting its vital importance.

b) Hepatic Portal Circulation (Nutrient-rich blood from digestive organs to liver)

The **hepatic portal circulation** is a specialized part of systemic circulation that connects the digestive tract directly to the liver. Unlike most veins, which return blood straight to the heart, the **hepatic portal vein** carries blood first to the liver for processing.

Pathway

1. **Collection of Nutrient-Rich Blood**
   - After digestion and absorption in the stomach, intestines, spleen, and pancreas, blood becomes rich in **glucose, amino acids, vitamins, and other absorbed nutrients**, but is relatively **low in oxygen.**
   - This blood is collected into the **hepatic portal vein**, the central vessel of the portal system.
2. **Delivery to the Liver**
   - The hepatic portal vein transports this nutrient-rich blood to the **liver sinusoids** (specialized capillaries in the liver).
   - Within the liver, hepatocytes (liver cells) perform multiple functions:
     - **Storage** of glucose as glycogen.
     - **Conversion** of excess amino acids and carbohydrates.
     - **Detoxification** of harmful substances (e.g., drugs, alcohol, toxins).
     - **Regulation** of nutrient levels before they enter the general circulation.
3. **Return to Systemic Circulation**
   - After processing, the cleansed blood leaves the liver via the **hepatic veins,** which drain directly into the **inferior vena cava.**
   - From here, the blood re-enters the systemic circuit, ready to be distributed to body tissues.

Key Function

The hepatic portal circulation acts as a **biological checkpoint.** It ensures that absorbed nutrients are properly balanced, harmful substances are detoxified, and metabolic homeostasis is maintained before the blood is allowed to circulate freely throughout the body.

c) Renal Circulation (Filtration of blood through kidneys)

The **renal circulation** is a specialized pathway of systemic circulation that ensures the kidneys receive a large volume of blood for filtration and homeostasis. About **20–25% of cardiac output** is directed to the kidneys, highlighting their crucial role in blood purification and fluid balance.

Pathway

1. **Entry of Oxygenated Blood**
   - Blood enters each kidney through a **renal artery,** a direct branch of the **abdominal aorta.**
   - The renal artery further divides into smaller arterioles, eventually reaching the **afferent arterioles** of millions of nephrons (the functional units of the kidney).

2. **Filtration in Nephrons**
   - In the nephron, the afferent arteriole leads to a capillary knot called the **glomerulus**, where blood plasma is filtered under pressure.
   - Filtrate (water, salts, glucose, urea, etc.) enters Bowman's capsule, while larger molecules like blood cells and proteins remain in circulation.
   - The blood then passes into the **efferent arteriole** and through a second capillary network (peritubular capillaries), where **reabsorption** (of water, salts, glucose) and **secretion** (of additional wastes) occur.
3. **Exit of Cleaned Blood**
   - After filtration and exchange, the blood — now deoxygenated but free from most metabolic wastes — is collected into the **renal vein**.
   - The renal vein drains into the **inferior vena cava**, returning cleansed blood to the systemic circulation.

Key Function

Renal circulation ensures constant internal balance (homeostasis) by:

- Removing nitrogenous wastes (urea, uric acid, creatinine).
- Regulating water and electrolyte levels.
- Maintaining blood pH and osmotic balance.
- Supporting blood pressure regulation through hormone secretion (e.g., renin, erythropoietin).

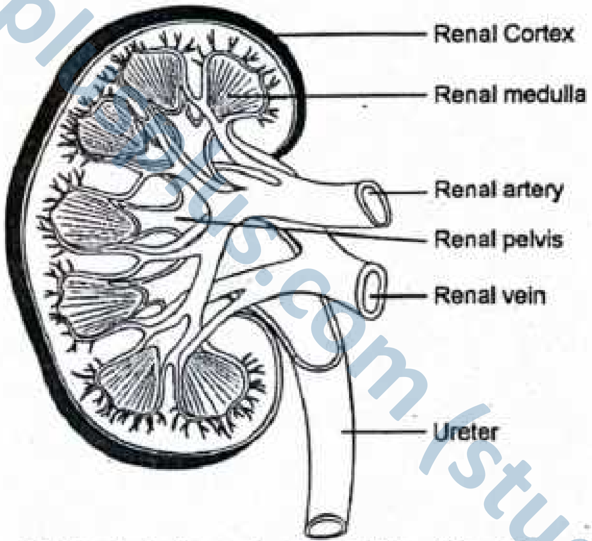


Figure: Principal arteries and veins of kidney (Renal Circulation)

► **Circulatory Pathway in Human Body**

Circulatory Type	Starting Point	Main Vessels Involved	Target/Organ(s)	Return Pathway
Pulmonary	Right ventricle	Pulmonary trunk → Pulmonary arteries	Lungs (for gas exchange)	Pulmonary veins → Left atrium
Systemic	Left ventricle	Aorta → Systemic arteries	All body tissues (except lungs)	Systemic veins → Venae cavae → Right atrium
Coronary	Aorta (base)	Right & left coronary arteries	Heart muscle (myocardium)	Cardiac veins → Coronary sinus → Right atrium
Hepatic Portal	Capillaries of gut organs	Hepatic portal vein	Liver (for processing absorbed substances)	Hepatic veins → Inferior vena cava
Renal	Abdominal aorta	Renal artery → Arterioles	Kidneys (for filtration)	Renal vein → Inferior vena cava

SLOs Based (MCQs)

CIRCULATORY PATHWAY IN HUMAN BODY

1. Which blood vessels carry oxygenated blood *to* the heart?

A. Pulmonary arteries B. Pulmonary veins
C. Coronary arteries D. Inferior vena cava

2. In which circulation does blood pass through two capillary networks before returning to the heart?

A. Pulmonary circulation B. Systemic circulation
C. Hepatic portal system D. Renal circulation

3. Which of the following structures receives deoxygenated blood directly from the myocardium?

A. Right ventricle B. Left atrium
C. Coronary sinus D. Aorta

4. What is the main artery that initiates systemic circulation?

A. Pulmonary artery B. Coronary artery
C. Hepatic artery D. Aorta

5. Which vessels supply blood to the kidneys?

A. Renal veins B. Afferent arterioles
C. Renal arteries D. Vasa recta

Answer Key

1. B	2. C	3. C	4. D	5. C

Short Questions

Q.1 What is pulmonary circulation?

Ans. Pulmonary circulation is the movement of blood between the heart and the lungs. Deoxygenated blood from the right ventricle is pumped into the pulmonary trunk, which divides into the right and left pulmonary arteries leading to the respective lungs. Inside the lungs, these arteries branch into arterioles and capillaries surrounding alveoli, where carbon dioxide is released and oxygen is absorbed. The now oxygenated blood flows into venules and then into four pulmonary veins, which return it to the left atrium of the heart. This circulation ensures proper oxygenation of blood and removal of carbon dioxide before it enters systemic circulation.

> **MARKING SCHEME (3 Marks)**
> - Definition – 1 mark
> - Pathway – 2 marks

Q.2 How does systemic circulation differ from pulmonary circulation?

Ans. Systemic circulation refers to the pathway in which oxygen-rich blood is pumped by the left ventricle through the aorta to supply all body tissues. Arteries branch into arterioles and capillaries, allowing exchange of nutrients, oxygen, and wastes. Deoxygenated blood is then collected by venules, veins, and finally the superior and inferior venae cavae, which empty into the right atrium. In contrast, pulmonary circulation involves only the heart and lungs. The right ventricle pumps deoxygenated blood through pulmonary arteries to the lungs for gas exchange, and oxygenated blood returns to the left atrium via pulmonary veins. Thus, systemic circulation serves the whole body, while pulmonary circulation focuses on gas exchange in the lungs.

> **MARKING SCHEME (3 Marks)**
> - Pathway of systemic circulation – 1.5 marks
> - Pathway of pulmonary circulation – 1.5 marks

Q.3 What is the function of the coronary circulation?

Ans. Coronary circulation is responsible for supplying blood to the heart muscle (myocardium), ensuring it receives the oxygen and nutrients required for continuous pumping. Coronary arteries branch directly from the ascending aorta soon after it leaves the left ventricle. The right and left coronary arteries further branch into smaller arteries to deliver oxygenated blood throughout the myocardium. Deoxygenated blood from heart tissue is collected by cardiac veins, which drain into the coronary sinus. The coronary sinus then opens into the right atrium, completing the pathway. Since the heart cannot rely on blood within its chambers for nourishment, this independent circulation is essential to maintain cardiac function and efficiency.

> **MARKING SCHEME (3 Marks)**
> - Function – 1 mark
> - Pathway – 2 marks

Q.4 Describe the hepatic portal system.

Ans. The hepatic portal system is a specialized vascular pathway that transports nutrient-rich blood from digestive organs to the liver for processing. Blood from the stomach, intestines, pancreas, and spleen flows into veins that merge to form the hepatic portal vein. This vein

> **MARKING SCHEME (3 Marks)**
> - Function – 1 mark
> - Pathway – 2 marks

carries blood into the liver, where it passes through hepatic sinusoids. In the liver, nutrients are metabolized, toxins are detoxified, and glucose is stored as glycogen. After processing, the blood is collected by central veins, which join to form hepatic veins. These hepatic veins drain directly into the inferior vena cava, returning blood to systemic circulation. This system ensures nutrients absorbed from food are processed before reaching the rest of the body.

Q.5 What are the main vessels involved in renal circulation?

Ans. Renal circulation refers to the blood supply of the kidneys. Oxygenated blood enters each kidney via the renal artery, which branches from the abdominal aorta. Inside the kidney, the renal artery divides into smaller arteries, leading to afferent arterioles that supply blood to the glomeruli for filtration. From the glomerulus, blood exits through efferent arterioles, which form two capillary networks: peritubular capillaries around the proximal and distal tubules, and the vasa recta alongside the loop of Henle. These networks facilitate reabsorption and secretion processes. Finally, blood flows into venules, merges into the renal vein, and drains into the inferior vena cava. This pathway is crucial for maintaining fluid, electrolyte, and waste balance.

MARKING SCHEME (3 Marks)
- Correct names of the vessels – 1 mark
- Pathway – 2 marks

SLO [B-12-R-46]
Compare the rate of blood flow through arteries, arterioles capillaries, venules and veins.

RATE OF BLOOD FLOW IN BLOOD VESSELS

Blood flow means simply the quantity of blood that passes through a given point in the circulation in a given period. The overall blood flow in the circulation of an adult at rest is about 5000 ml/min. This is called cardiac output. It is the amount of blood pumped by the heart in a unit period.

Comparison of the Rate of Blood Flow through Arteries, Arterioles, Capillaries, Venules and Veins

Blood travels over a thousand times faster in the aorta, i.e., about 30 cm/sec on average than in capillaries i.e., about 0.26 cm/sec. You might think that blood should travel faster through capillaries than through arteries, because the diameter of capillaries is very small. However, it is the total cross-sectional area of capillaries that determines flow rate. Each artery conveys blood to such an enormous number of capillaries that the total cross-sectional area is much greater in capillary beds than in any other part of the circulatory system. For this reason the blood slows substantially as it enters the arterioles from arteries and slows further still in the capillary beds. As blood leaves the capillaries and enters the venules and veins it speeds up again as a result of the reduction in total cross-sectional area. The carotid sinus and aortic arch baroreceptor reflexes are important in regulating blood pressure moment to moment.

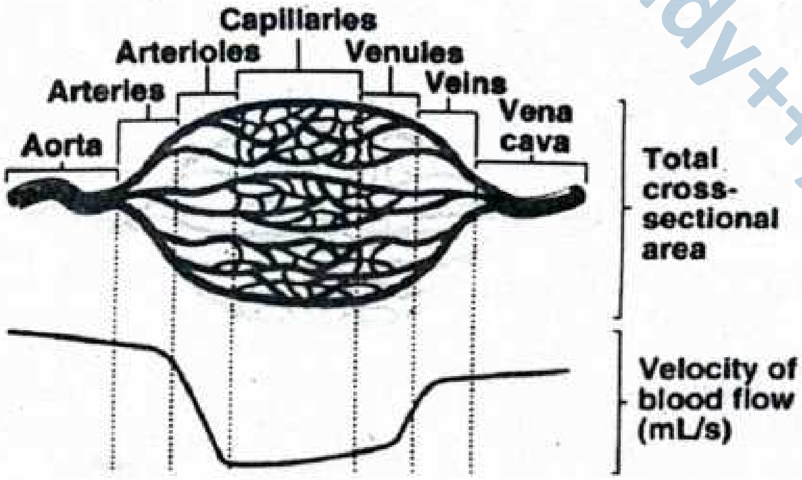


Figure: Blood vessel types and velocity of blood flow: Total cross-sectional area for each of the major blood vessel types is the space through which blood flows, measured in square centimeters. The cross-sectional area of the aorta is about 2.5 cm^2. The crosssectional area of each capillary is much smaller, but there are so many that the total cross-sectional area is more than that of the aorta. The line at the bottom of the graph shows that blood velocity drops dramatically in arterioles, capillaries, and venules. As the total cross-sectional area increases the velocity of blood flow decreases.

Feature	Pulmonary Artery	Pulmonary Vein
Direction of Flow	Carries blood **away** from the heart	Carries blood **towards** the heart
Type of Blood	Carries **deoxygenated** blood	Carries **oxygenated** blood
Connected to	Right Ventricle	Left Atrium
Number of Vessels	One, which branches into two arteries	Four veins (two from each lung)
Destination or Origin	Takes blood to the **lungs**	Brings blood from the **lungs**

SLO [B-12-R-47]
Define blood pressure and explain its periods of systolic and diastolic pressure.

BLOOD PRESSURE

Blood pressure is the **force exerted by circulating blood on the walls of blood vessels**, especially the arteries. It results from the pumping action of the heart and the resistance offered by the arterial walls. Blood pressure is an essential indicator of cardiovascular health and ensures that blood reaches all tissues with sufficient force for the exchange of oxygen, nutrients, and wastes.

▶ **Periods of Blood Pressure:**

1. **Systolic Pressure (Higher Value)**
   - This is the **maximum arterial pressure** reached during **ventricular systole**, the phase when the ventricles contract forcefully and pump blood into the aorta and pulmonary trunk.
   - It directly reflects the **strength of the heart's contraction** and the efficiency of blood ejection into the arterial system.
   - In a healthy adult at rest, the systolic pressure averages around **120 mmHg**.
   - It provides an indication of how hard the heart must work to circulate blood throughout the body.
2. **Diastolic Pressure (Lower Value)**
   - This is the **minimum arterial pressure** recorded during **ventricular diastole**, the phase when the ventricles relax and refill with blood.
   - It depends largely on the **elastic recoil of the arterial walls**, which maintains continuous blood flow even when the heart is not actively contracting.
   - In a healthy adult at rest, the diastolic pressure averages around **80 mmHg**.
   - It represents the **baseline pressure** required to keep blood moving steadily through the tissues between heartbeats.

▶ **Normal Range and Expression:**

Blood pressure is always expressed as a **fraction** with two values:
- **Systolic pressure** (upper number) – represents the maximum pressure during ventricular contraction.
- **Diastolic pressure** (lower number) – represents the minimum pressure during ventricular relaxation.

For example, a typical reading is written as **120/80 mmHg.**
- Here, **120 mmHg** is the systolic pressure.
- **80 mmHg** is the diastolic pressure.

The unit **mmHg** (millimeters of mercury) is used because mercury columns were traditionally employed in sphygmomanometers (blood pressure measuring instruments).

▶ ***Normal Adult Range:***
- Systolic pressure: 110–130 mmHg
- Diastolic pressure: 70–85 mmHg

This balanced range ensures that enough pressure is generated to deliver blood efficiently without overburdening the heart or damaging blood vessels.
Systolic pressure shows the **pumping strength of the heart**, while diastolic pressure reflects the **baseline pressure** that maintains continuous blood flow. Both are vital for normal tissue perfusion.

SLO [B-12-R-48]
State the role of baroreceptors and volume receptor in regulating the blood pressure.

BARORECEPTORS AND VOLUME RECEPTORS

Baroreceptors are specialized **mechanoreceptors** that monitor changes in blood pressure and relay signals to the brain to initiate appropriate regulatory responses. They function through a **negative feedback mechanism** known as the **baroreflex.**

1. High-Pressure Arterial Baroreceptors

Located in the **walls of the aorta** and the **carotid sinus**, these receptors detect **changes in arterial blood pressure.** When blood pressure rises:

- **Parasympathetic** stimulation is increased.
- **Vasodilation** occurs throughout the body.
- **Heart rate slows,** and **blood pressure falls.**

Conversely, a drop in blood pressure activates the **sympathetic nervous system**, leading to:

- **Vasoconstriction,**
- **Increased** heart rate, and
- **Restoration** of normal blood pressure.

2. Low-Pressure Baroreceptors (Volume Receptors)

These are found in the **atria of the heart** and **carotid arteries**, where they monitor **blood volume.** A decrease in blood volume triggers:

- A signal to the **hypothalamus,**
- Increased release of **antidiuretic hormone (ADH),**
- **Water reabsorption** in the kidneys,
- **Restoration of blood volume and pressure.**

SLOs Based (MCQs)

BLOOD PRESSURE AND BARORECEPTORS

1. **Which of the following blood vessels has the slowest rate of blood flow?**
   A. Aorta B. Arterioles
   C. Capillaries D. Veins
2. **What causes the increase in blood velocity as blood flows from capillaries into veins?**
   A. Increase in vessel diameter
   B. Increase in total cross-sectional area
   C. Decrease in total cross-sectional area
   D. Vasodilation
3. **Systolic pressure is measured during:**
   A. Atrial contraction B. Atrial relaxation
   C. Ventricular contraction D. Ventricular relaxation
4. **Which hormone is released in response to low-pressure baroreceptor stimulation?**
   A. Adrenaline
   B. Insulin
   C. Antidiuretic hormone (ADH)
   D. Glucagon
5. **High-pressure baroreceptors are primarily located in:**
   A. Pulmonary veins and atria
   B. Carotid sinus and aorta
   C. Renal arteries
   D. Jugular vein and vena cava

Answer Key

1. C	2. C	3. C	4. C	5. B

Short Questions

Q.1 What is cardiac output and how is it related to blood flow?

Ans. Cardiac output is defined as the volume of blood pumped by each ventricle of the heart per minute. In a healthy resting adult, it averages about 5000 mL/min, though it can increase several times during exercise. Cardiac output is calculated as the product of heart rate and stroke volume, making it a direct indicator of how effectively the heart meets the body's metabolic needs. Since it represents the total blood flow through the circulatory system, it is vital for ensuring proper delivery of oxygen and nutrients to tissues, as well as removal of metabolic wastes. Thus, cardiac output reflects overall cardiovascular performance and systemic blood flow.

> **MARKING SCHEME (3 Marks)**
> - Definition – 1 mark
> - Function – 2 marks

Q.2 Why does blood flow slow down in capillaries despite their narrow diameter?

Ans. Although individual capillaries are extremely narrow, their immense number throughout the body creates a total cross-sectional area far greater than any other section of the circulatory system. According to the principle of flow dynamics, when total cross-sectional area increases, the velocity of blood decreases. This explains why blood flows much more slowly in capillaries compared to arteries or veins. The reduced flow is not a limitation but an adaptation: it provides sufficient time for efficient exchange of gases, nutrients, and wastes between blood and tissues. Therefore, the structural arrangement of capillaries ensures that every cell receives proper nourishment and metabolic balance is maintained.

> **MARKING SCHEME (3 Marks)**
> - Structure/total cross-sectional area – 2 marks
> - Function – 1 mark

Q.3 Define blood pressure and explain the terms systolic and diastolic pressure.

Ans. Blood pressure is the force exerted by circulating blood against the walls of blood vessels, particularly arteries. It is one of the most important indicators of cardiovascular health and is measured in millimeters of mercury (mmHg). The higher reading, called **systolic pressure**, represents the peak pressure in the arteries when the ventricles contract and eject blood. The lower reading, known as **diastolic pressure**, occurs when the ventricles relax and the heart refills with blood. A typical normal blood pressure in adults is around 120/80 mmHg, where 120 denotes systolic and 80 represents diastolic pressure. Both values are crucial for assessing cardiac function and arterial health.

> **MARKING SCHEME (3 Marks)**
> - Systolic pressure definition – 1.5 marks
> - Diastolic pressure definition – 1.5 marks

Q.4 How do high-pressure baroreceptors regulate blood pressure?

Ans. High-pressure baroreceptors are specialized stretch receptors located in the aortic arch and the carotid sinus. They monitor arterial blood pressure by detecting changes in the degree of vessel wall stretch. When blood pressure rises, baroreceptors send increased signals to the cardiovascular center in the medulla, enhancing parasympathetic activity. This slows heart rate (bradycardia) and causes vasodilation, lowering arterial pressure. Conversely, when blood pressure drops, they reduce their firing rate, stimulating sympathetic nerves. This leads to vasoconstriction, increased heart rate, and stronger cardiac contractions, thereby raising blood pressure back to normal. Thus, baroreceptors provide a rapid, short-term mechanism for maintaining blood pressure stability.

> **MARKING SCHEME (3 Marks)**
> - High-pressure baroreceptor function – 1 mark
> - Mechanism – 2 marks

Q.5 Describe the role of low-pressure baroreceptors (volume receptors).

Ans. Low-pressure baroreceptors, also known as volume receptors, are located primarily in the atria and pulmonary veins. Unlike high-pressure baroreceptors, they do not sense arterial pressure directly but monitor changes in blood volume through stretch detection. When blood volume decreases, these receptors send signals to the hypothalamus, which stimulates the release of antidiuretic hormone (ADH) from the posterior pituitary. ADH acts on the kidneys to promote water reabsorption, thereby conserving fluid and increasing blood volume. Conversely, when blood volume is high, ADH release is suppressed, leading to increased urine output. In this way, low-pressure baroreceptors help maintain circulatory volume and contribute to long-term regulation of blood pressure.

> **MARKING SCHEME (3 Marks)**
> - Low-pressure baroreceptor function – 1 mark
> - Mechanism – 2 marks

SLO [B-12-R-49]
Define the term thrombus and differentiate between thrombus and embolus.

6

THROMBUS AND EMBOLUS

A **thrombus** is an **abnormal, solid mass of clotted blood** that forms inside an **unbroken blood vessel or within the heart chambers**. It develops when the normal process of blood clotting (coagulation) occurs inappropriately, without any external injury.

- **Composition:** A thrombus is mainly composed of platelets, a network of fibrin protein threads, and trapped red and white blood cells.
- **Difference from normal clot:** In contrast to a normal clot (formed on a wound to prevent blood loss), a thrombus forms within intact vessels, where it is undesirable and harmful.
- **Blocking effect:** Depending on its size and location, a thrombus may narrow (partial obstruction) or completely block (total obstruction) the flow of blood in the vessel. This reduces oxygen supply to tissues and may result in serious complications such as tissue death (infarction).
- **Sites of formation:** Thrombi are commonly found in arteries, veins, or the heart. For example, a thrombus in a coronary artery may lead to a myocardial infarction (heart attack), while a thrombus in deep veins of the leg may lead to deep vein thrombosis (DVT).

► **Difference between Thrombus and Embolus**

Feature	Thrombus	Embolus
Definition	A stationary blood clot formed within a vessel.	A detached piece of a clot (or other material like fat, air, or bacteria) that travels in the bloodstream.
Position	Fixed at the site of formation.	Moves with blood flow until it lodges elsewhere.
Effect	Causes local obstruction, reducing or stopping blood supply at that spot.	Causes sudden blockage in a distant vessel (e.g., brain, lungs, kidney).
Example	Coronary thrombus → blocks a coronary artery, leading to a heart attack.	Pulmonary embolus → a clot traveling from leg veins to lungs, causing breathing difficulty.

SLO [B-12-R-50]
Identify the factors causing atherosclerosis and arteriosclerosis.

ATHEROSCLEROSIS AND ARTERIOSCLEROSIS

- **Atherosclerosis** is a specific type of arterial disease in which **fatty deposits, called plaques**, accumulate inside the inner lining of arteries.
  - These plaques are made of cholesterol, fats, calcium, and cellular waste products.
  - As plaques grow, they **narrow the arterial lumen**, reducing blood flow.
  - In severe cases, the plaque may rupture, causing clot formation and complete blockage of the artery.
- Arteriosclerosis is a broader condition that refers to the hardening, thickening, and loss of elasticity of arterial walls.
  - It usually develops gradually with **aging** as arteries become stiffer.
  - Atherosclerosis is considered one form of arteriosclerosis, but other types may occur due to calcium deposition or long-term high blood pressure.

► **Impact on Blood Circulation**

Both conditions *seriously impair* the efficiency of blood circulation:

- Reduced blood supply to vital organs such as the heart, brain, and kidneys.
- Increased workload on the heart, which must pump harder against narrowed, stiff arteries.
- Risk of life-threatening events, including heart attack (myocardial infarction) and stroke (cerebrovascular accident), when blood supply to the heart or brain is suddenly blocked.

▶ Factors Causing Atherosclerosis

Atherosclerosis develops gradually due to a combination of lifestyle, genetic, and health-related factors. The following are the **major contributors:**

1. **High Blood Cholesterol**
   - Excess **low-density lipoproteins (LDL, "bad cholesterol")** seep into artery walls and form fatty streaks.
   - Over time, these deposits harden into plaques, narrowing the arterial lumen.
2. **High Blood Pressure (Hypertension)**
   - Constant high pressure damages the **endothelium (inner lining of arteries).**
   - This makes the vessel wall more vulnerable to cholesterol deposits and **plaque buildup.**
3. **Smoking**
   - Chemicals like **nicotine and carbon monoxide** injure blood vessel walls.
   - Smoking also increases LDL, decreases **high-density lipoproteins (HDL, "good cholesterol"),** and makes blood more prone to clotting.
4. **Diabetes Mellitus**
   - Persistently high blood sugar levels cause **glycation damage** to vessel walls.
   - This accelerates fatty plaque formation and doubles the risk of atherosclerosis.
5. **Obesity and Unhealthy Diet**
   - Diets rich in **saturated fats, trans fats, cholesterol, and refined sugars** raise blood lipids.
   - Obesity also promotes hypertension, diabetes, and metabolic syndrome — all risk factors for atherosclerosis.
6. **Sedentary Lifestyle**
   - *Lack* of physical activity lowers **HDL ("good cholesterol")** levels.
   - Physical inactivity also promotes obesity, high blood pressure, and poor circulation.
7. **Family History and Genetics**
   - Inherited conditions like **familial hypercholesterolemia** cause extremely high LDL levels.
   - A positive family history of early heart disease increases susceptibility.
8. **Chronic Inflammation**
   - Long-term infections, autoimmune diseases (e.g., rheumatoid arthritis), or chronic inflammatory states damage arterial walls.
   - This ongoing injury provides a site for plaques to form more easily.

▶ Factors Causing Arteriosclerosis

Arteriosclerosis refers to the **general hardening, thickening, and loss of elasticity** of arterial walls. Unlike atherosclerosis (which is due to fatty plaque buildup), arteriosclerosis can occur even without lipid deposits, especially with aging. The main contributing factors are:

1. **Aging**
   - With age, arteries naturally become **less elastic and more rigid.**
   - This process reduces the ability of blood vessels to expand and contract with each heartbeat, leading to increased blood pressure.
2. **Long-term Hypertension (High Blood Pressure)**
   - Persistent high pressure against arterial walls causes **structural changes.**
   - Arteries respond by thickening and stiffening, which further increases resistance to blood flow.
3. **Calcium Deposition**
   - Over time, **minerals like calcium** accumulate in the middle layer of arteries.
   - This calcification makes arterial walls brittle and less flexible (a process often referred to as "hardening of the arteries").
4. ***Atherosclerosis***
   - Although distinct, atherosclerosis can contribute to arteriosclerosis.
   - Fatty plaques within the arteries promote **scarring and stiffening,** worsening the loss of elasticity.
5. **Lifestyle Risks**
   - Many of the same factors that cause atherosclerosis also worsen arteriosclerosis:

- **Smoking** damages the endothelium.
- **Obesity and poor diet** promote vascular stress.
- **Physical inactivity** accelerates arterial stiffening.

SLO [B-12-R-51]
Categorize Angina pectoris, heart attack, and heart failure as the stages of cardiovascular disease development.

STAGES OF CARDIOVASCULAR DISEASE DEVELOPMENT

Cardiovascular diseases develop gradually, often starting with reduced blood flow to the heart muscle and progressing to more serious complications. The three main clinical conditions are:

1. Angina Pectoris (Warning Stage)

Definition:

Angina **pectoris** is a clinical condition characterized by sudden **chest pain** or discomfort caused by a **temporary and reversible** reduction of blood flow to the myocardium. This usually results from partial narrowing of coronary arteries due to **atherosclerotic plaques** or from coronary artery spasms. **Unlike a heart attack**, the myocardial cells do not die, but they suffer from **oxygen deficiency** (ischemia).

Features (Clinical Presentation):

- Pain is commonly described as a **tightness, heaviness, or squeezing pressure** in the chest.
- Discomfort may radiate to the **left shoulder, arm, neck, or jaw.**
- Symptoms are usually **triggered by exertion, emotional stress, or heavy meals**, when the heart requires more oxygen.
- Pain typically subsides with **rest** or by taking vasodilator medicines such as **nitroglycerin**, which widen coronary arteries and restore blood flow.
- Attacks are usually short-lived (a few minutes).

Significance (Medical Importance):

- Angina serves as an important **warning sign** that the heart muscle is not receiving adequate oxygen.
- If the underlying blockage continues to worsen, it may progress to **myocardial infarction (heart attack)**, where cardiac muscle damage becomes permanent.
- Therefore, early recognition and management of angina are essential in preventing severe cardiovascular complications.

2. Heart Attack (Myocardial Infarction – Acute Stage)

Definition:

A **heart attack**, or **myocardial infarction (MI)**, occurs when there is a **complete and prolonged blockage of a coronary artery**, usually by a blood clot (thrombus) formed on top of an atherosclerotic plaque. As a result, the affected region of the myocardium is completely deprived of oxygen (**ischemia**), leading to **permanent injury and death of cardiac muscle cells (necrosis)**.

Features (Clinical Presentation):

- Severe, **crushing**, or burning chest pain that persists for more than 20 minutes and is not **relieved** by rest or **nitroglycerin**.
- **Pain often radiates** to the left arm, shoulder, neck, jaw, or back.
- Associated symptoms include **profuse sweating** (cold sweat), shortness of breath, palpitations, dizziness, and nausea/vomiting.
- In some cases (*especially* in elderly and diabetic patients), a **"silent heart attack"** may occur with minimal or atypical symptoms.

Significance (Medical Importance):

- Myocardial infarction is a **medical emergency**. Without immediate intervention (such as clot-dissolving drugs, angioplasty, *or bypass* surgery), the damage to heart tissue becomes irreversible.
- Loss of functioning cardiac muscle weakens **the pumping ability of the heart**, predisposing to **arrhythmias, heart failure, or sudden death**.
- Rapid diagnosis and treatment are critical for patient survival and long-term recovery.

3. Heart Failure (Chronic Stage)

Definition:

Heart failure is a **chronic and progressive condition** in which the heart becomes **too weak or too stiff to pump blood efficiently**, failing to meet the body's oxygen and nutrient demands. Unlike a heart attack (sudden event), heart failure develops **gradually** over months or years as the heart muscle sustains cumulative damage.

Causes:

- **Repeated heart attacks (myocardial infarctions):** Permanent loss of cardiac muscle reduces pumping power.
- **Long-term uncontrolled hypertension:** Increases the heart's workload, causing thickening (hypertrophy) followed by weakening.
- **Valvular heart disease, cardiomyopathy, or chronic arrhythmias** may also contribute.

Features (Clinical Presentation):

- **Persistent fatigue and weakness,** leading to reduced ability to perform daily activities.
- **Shortness of breath (dyspnea):** Worsens on exertion or when lying flat (**orthopnea**).
- **Fluid retention:** Swelling (**edema**) in ankles, feet, and legs; congestion of lungs causes **cough and difficulty breathing at night.**
- **Rapid weight gain** due to fluid accumulation.
- *In severe cases,* fluid may also accumulate in the abdomen (**ascites**).

Significance (Medical Importance):

- Heart failure represents the **final stage of cardiovascular disease progression** (after angina and heart attack).
- It leads to a **chronic decline in quality of life,** frequent hospitalizations, and increased risk of premature death.
- Treatment (medications, lifestyle changes, surgery, or devices like pacemakers) aims to **control symptoms, prevent worsening, and improve survival,** but complete recovery is rare.

SLO [B-12-R-52]
State the congenital heart problem related to the malfunction of cardiac valves.

CONGENITAL HEART PROBLEM RELATED TO CARDIAC VALVES

A congenital heart problem means a **structural defect present at birth.** When it involves the **cardiac valves,** it affects the normal opening and closing of valves, disrupting blood flow through the heart.

Types of Valve Malfunctions:

1. Valve Stenosis:

Valve stenosis is a congenital heart defect in which the opening of a cardiac valve becomes **abnormally narrow.** This narrowing usually results from **thickened, stiff, or fused valve leaflets** that fail to open fully. Because of the restricted opening, **forward blood flow is obstructed,** and the heart must **generate higher pressure** to push blood through the narrowed valve. Over time, this extra workload causes **hypertrophy (thickening)** of the heart chamber located behind the affected valve.

Examples:

- **Pulmonary Stenosis** – narrowing at the pulmonary valve, reducing blood flow from the right ventricle to the lungs.
- **Aortic Stenosis** – narrowing at the aortic valve, restricting blood flow from the left ventricle to the aorta and systemic circulation.

Valve stenosis can lead to fatigue, shortness of breath, chest pain, fainting, and, if untreated, may progress to heart failure.

2. Valve Regurgitation *(or Incompetence)*:

Valve regurgitation is a congenital heart defect in which a cardiac valve **fails to close completely.** As a result, blood flows **backward (backflow)** instead of moving forward in one direction.

This backflow of blood leads to **volume overload** in the affected heart chambers. Over time, the chambers enlarge (dilate) and the heart muscle weakens because it must pump the same blood repeatedly.

Example:

- **Mitral Valve Regurgitation** – the mitral valve does not close properly, causing blood to flow backward from the left ventricle into the left atrium during ventricular contraction.

Valve regurgitation reduces the efficiency of blood circulation, leading to symptoms such as fatigue, shortness of breath, palpitations, and in severe cases, heart failure.

Common Congenital Valve Defects:

Congenital valve defects are structural abnormalities of the cardiac valves present at birth. These defects disturb the normal one-way flow of blood, often leading to stenosis (narrowing) or regurgitation (backflow).

1. **Bicuspid Aortic Valve (BAV)**

Description:

The aortic valve normally has **three cusps (tricuspid), but in this condition, it is formed with only two cusps (bicuspid).**

Pathology:

- The abnormal structure causes the valve to **wear out more quickly** than normal.
- This increases the risk of **aortic stenosis** (narrowing of the valve opening) or **aortic regurgitation** (backflow of blood into the left ventricle).

Clinical Features:

- May remain **silent in childhood**, but symptoms such as chest pain, dizziness, or shortness of breath often appear in adulthood.
- Increases risk of **infective endocarditis** (valve infection).

Significance:

It is the **most common congenital valve defect** and may require surgical repair or valve replacement later in life.

2. **Congenital Pulmonary Stenosis**

Description:

The **pulmonary valve**, located between the right ventricle and pulmonary artery, is abnormally **narrow or thickened.**

Pathology:

- Obstruction restricts blood flow from the right ventricle to the lungs.
- This forces the right ventricle to pump harder, leading to **hypertrophy (thickening)** of the ventricular wall.

Clinical Features:

- In mild cases, children may remain asymptomatic.
- Severe cases cause **cyanosis (bluish skin due to lack of oxygen)**, fatigue, shortness of breath, and delayed growth.

Significance:

Early detection is crucial; treatment may involve **balloon valvuloplasty** or surgery to relieve the obstruction. Both conditions interfere with normal cardiac efficiency and, if left untreated, may progress to **heart failure or life-threatening complications.**

Consequences:

- Fatigue, shortness of breath, poor growth in children.
- Cyanosis (bluish skin) in severe defects due to poor oxygenation.
- If untreated, can lead to **heart failure** or **reduced life expectancy.**

SLO [B-12-R-53]
Describe the principles of angiography.

ANGIOGRAPHY – PRINCIPLES AND DESCRIPTION

Angiography is a **diagnostic imaging technique** used to visualize the inside of blood vessels and the chambers of the heart. It helps detect blockages, narrowing, malformations, or other abnormalities of the circulatory system.

Principles of Angiography

1. **Contrast Medium Injection**
   - A special dye (contrast medium) is injected into the bloodstream through a catheter.
   - This dye absorbs **X-rays**, making blood vessels visible on the radiograph.
2. **Catheterization**
   - A thin, flexible tube (catheter) is inserted into a large artery (commonly the femoral artery in the groin or radial artery in the arm).
   - The catheter is guided to the target blood vessel (e.g., coronary arteries, cerebral arteries).
3. **X-ray Imaging (Fluoroscopy)**
   - Continuous X-rays (fluoroscopy) capture real-time images as the contrast flows through the vessels.
   - Blockages, leakages, or abnormal vessel structures appear as interruptions or narrowing in the image.
4. **Recording and Analysis**
   - Images are stored as angiograms, which allow physicians to **evaluate circulation, identify diseases, and plan treatment** (e.g., angioplasty, stenting, or surgery).

Types of Angiography

1. **Coronary Angiography**
   - **Focus:** Coronary arteries that supply oxygenated blood to the heart muscle.
   - **Purpose:** Detects blockages, narrowing (stenosis), or complete obstruction that may lead to **angina pectoris** or **myocardial infarction (heart attack).**
   - **Clinical Use:** Essential in diagnosing **coronary artery disease (CAD)** and planning interventions such as **angioplasty or bypass surgery.**
2. **Cerebral Angiography**
   - **Focus:** Arteries and veins in the **brain.**
   - **Purpose:** Identifies **aneurysms, arteriovenous malformations (AVMs), stroke-related blockages, or bleeding sources.**
   - **Clinical Use:** Guides neurosurgeons in treating strokes, brain tumors, or vascular malformations.
3. **Pulmonary Angiography**
   - **Focus: Pulmonary arteries** that carry blood from the heart to the lungs.
   - **Purpose:** Detects **pulmonary embolism (PE)**, where a blood clot obstructs blood flow to lung tissue.
   - **Clinical Use:** Helps confirm PE diagnosis when other tests (like CT scan or ultrasound) are inconclusive.
4. **Peripheral Angiography**
   - **Focus:** Blood vessels of the **arms and legs.**
   - **Purpose:** Identifies **narrowing, blockages, or aneurysms** in limb circulation.
   - **Clinical Use:** Commonly used for diagnosing **peripheral arterial disease (PAD)**, which can cause leg pain, ulcers, or gangrene if untreated.

Clinical Applications

1. **Diagnosis of Coronary Artery Disease (CAD)**
   - Angiography is the **gold standard** for detecting blockages or narrowing in the coronary arteries.
   - It helps determine the **severity** and **exact location** of the obstruction that may cause **angina** or **myocardial infarction.**
2. **Detection of Aneurysms, Vascular Malformations, or Tumours**
   - By providing a clear image of blood vessels, angiography can identify **aneurysms (abnormal dilations), arteriovenous malformations (AVMs), or tumours** that receive an abnormal blood supply.
   - This is especially useful in the **brain, lungs, and kidneys**, where early diagnosis prevents life-threatening complications.
3. ***Evaluation before Cardiac Surgery or Angioplasty***
   - Surgeons use angiographic results to plan coronary bypass surgery or angioplasty with stent placement.
   - It ensures precision by identifying which arteries are diseased and require intervention.

4. **Monitoring Post-Surgical Recovery of Stents or Grafts**
   - After angioplasty or bypass surgery, angiography helps monitor whether **stents remain open** or **grafts function properly.**
   - It also detects **restenosis (re-narrowing)** of arteries, guiding further treatment.

Risks and Limitations

Although angiography is a highly valuable diagnostic tool, it carries certain risks and limitations that students must understand.

▶ **Minor Risks**
- **Bleeding or Bruising:** May occur at the catheter insertion site (usually in the groin or wrist).
- **Infection:** Rare, but possible at the puncture site if proper sterilization is not maintained.
- **Allergic Reaction:** Some patients may develop itching, rash, or swelling due to sensitivity to the **iodine-based contrast dye.**

▶ **Rare but Serious Risks**
- **Kidney Damage:** The contrast dye may harm the kidneys, especially in patients with **pre-existing kidney disease or diabetes.**
- **Vascular Injury:** The catheter may damage blood vessel walls, leading to clot formation or, rarely, vessel rupture.
- **Heart Attack or Stroke:** Extremely rare, but possible if a blood clot dislodges during the procedure.

▶ **Limitations**
- **Invasiveness:** Angiography requires **catheter insertion into blood vessels,** making it more invasive than newer imaging methods.
- **Radiation Exposure:** Involves X-rays, which carry a small radiation risk.
- **Cost and Availability:** Angiography is expensive and requires specialized equipment and trained personnel.

▶ **Non-Invasive Alternatives**

To reduce risks, doctors may use:
- **CT Angiography (CTA):** Uses computed tomography with contrast dye, providing detailed 3D images of blood vessels without catheterization.
- **Magnetic Resonance Angiography (MRA):** Uses magnetic fields and radio waves (sometimes with dye) to visualize vessels without radiation.

> **SLO [B-12-R-54]**
> **Outline the main principle of coronary bypass, angioplasty and open-heart surgery.**

MAIN PRINCIPLES OF CORONARY BYPASS, ANGIOPLASTY, AND OPEN-HEART SURGERY

1. Coronary Artery Bypass Graft (CABG)

▶ **Principle:**

The principle of coronary artery bypass (CABG) is to restore an adequate supply of oxygen-rich blood to the heart muscle by creating an alternate route around blocked coronary arteries. In this procedure, a healthy blood vessel (graft), usually taken from the patient's leg vein (saphenous vein) or chest artery (internal mammary artery), is surgically attached above and *below the site of arterial blockage*. This allows blood to flow freely through the new pathway instead of being obstructed by the narrowed or clogged artery. By maintaining continuous blood circulation to the myocardium, the bypass relieves chest pain (angina), improves heart function, and reduces the risk of heart attack.

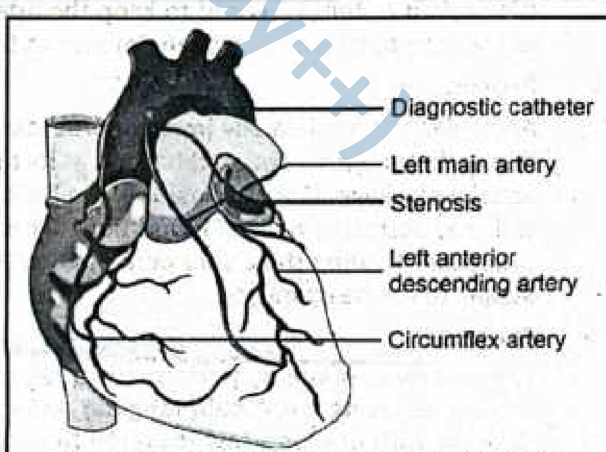


Figure: Coronary angiogram-schematic of the vessels and branches

▶ **Procedure:**

In coronary artery bypass surgery, a healthy blood vessel—commonly the saphenous vein from the leg or the internal mammary artery from the chest—is carefully harvested and grafted to bypass the blocked coronary artery. This new pathway allows oxygen-rich blood to flow around the obstruction and reach the heart muscle, restoring adequate oxygen supply. During the surgery, the heart may be temporarily stopped, and a heart-lung machine maintains circulation and oxygenation. The procedure can involve one or multiple grafts, depending on the number and location of blockages, and it is performed to relieve symptoms such as angina, improve cardiac function, and reduce the risk of heart attack.

▶ **Purpose:**

Coronary bypass surgery is primarily indicated when multiple or severe blockages obstruct blood flow through the coronary arteries. By creating alternative pathways for blood to reach the myocardium, the procedure significantly improves oxygen supply to the heart muscle. This not only relieves symptoms such as chest pain (angina) but also reduces the risk of life-threatening events like heart attacks. In addition, restoring adequate perfusion helps improve overall cardiac function, exercise tolerance, and quality of life for patients with advanced coronary artery disease.

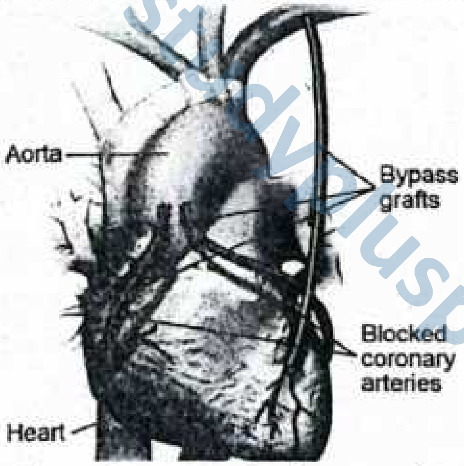


Figure: Coronary artery bypass graft.

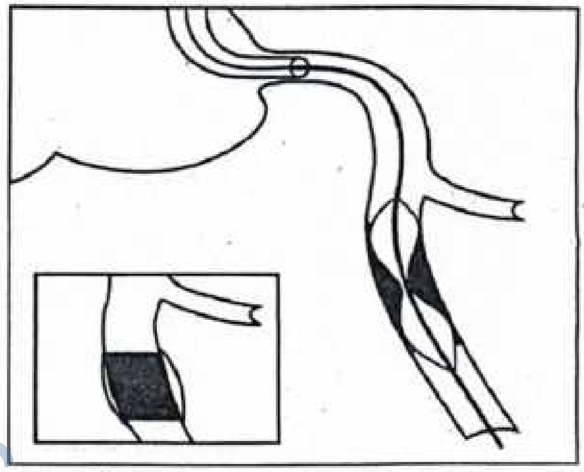


Figure: Coronary artery bypass graft.

2. Angioplasty (Percutaneous Coronary Intervention – PCI)

▶ **Principle:**

Angioplasty is a minimally invasive procedure designed to widen narrowed or blocked coronary arteries and restore normal blood flow to the heart muscle. A thin, flexible catheter with a deflated balloon at its tip is inserted into the affected artery and guided to the site of blockage. Once in position, the balloon is inflated, compressing the plaque against the arterial walls and enlarging the vessel lumen. Often, a small metal mesh tube called a stent is placed to keep the artery open permanently. This procedure relieves symptoms such as chest pain, improves oxygen delivery to the myocardium, and reduces the risk of heart attack.

▶ **Procedure:**

Angioplasty is a minimally invasive procedure designed to widen narrowed or blocked coronary arteries. A thin catheter with a deflated balloon at its tip is carefully guided through the blood vessels to the site of the arterial blockage. Once in position, the balloon is inflated, compressing the fatty plaque against the arterial wall and restoring normal blood flow. Often, a stent—a small metallic mesh tube—is inserted during the procedure to keep the artery permanently open, preventing re-narrowing and ensuring continuous oxygen supply to the heart muscle.

▶ **Purpose:**

Angioplasty is primarily performed to restore blood flow through narrowed or blocked coronary arteries rapidly, relieving chest pain (angina) and reducing the risk of a heart attack. It is especially suitable for patients with one or a few localized blockages, offering a less invasive alternative to open-heart surgery. Compared to traditional bypass surgery, angioplasty involves a shorter hospital stay and faster recovery, allowing patients to resume normal activities sooner while maintaining improved cardiac function.

3. Open-Heart Surgery

- **Principle:**

Open-heart surgery involves surgically opening the chest and temporarily stopping the heart to perform complex procedures on the heart's chambers, valves, or major vessels. A heart-lung machine maintains circulation and oxygenation of the blood during the operation. The principle of this surgery is to directly correct structural or functional abnormalities, such as valve replacement, septal defect repair, or coronary artery bypass, under controlled conditions. By allowing surgeons to operate on a still and bloodless heart, the procedure ensures precise repair or replacement, ultimately restoring normal cardiac function and improving patient survival.

- **Procedure:**

Open-heart surgery involves opening the sternum to gain direct access to the heart. To allow precise surgical intervention, the heart is temporarily stopped using specialised drugs, while a cardiopulmonary bypass machine takes over the functions of oxygenating and circulating the blood. This machine ensures that the body continues to receive oxygenated blood throughout the operation. Surgeons can then perform various corrective procedures, such as repairing malfunctioning valves, correcting congenital heart defects, or carrying out coronary artery bypass grafting (CABG) to restore proper blood flow to the myocardium.

- **Purpose:**

Open-heart surgery allows surgeons to directly visualize and access the heart, enabling precise correction of structural or functional defects. It is particularly employed in complex cases where minimally invasive techniques, such as angioplasty or catheter-based interventions, are insufficient. The procedure can repair malfunctioning valves, correct congenital anomalies, or restore adequate blood flow via coronary artery bypass grafting, ultimately improving cardiac function and patient outcomes.

SLOs Based (MCQs)

CARDIOVASCULAR DISORDERS, DIAGNOSIS AND TRAETMENTS

1. Which of the following is a common symptom of hypotension?

A. Chest pain
B. Dizziness upon standing
C. Severe headache
D. Palpitations

2. What is the normal adult blood pressure range considered by physicians?

A. 140/90 mmHg B. 100/60 mmHg
C. 120/80 mmHg D. 90/60 mmHg

3. Which diagnostic test provides a graphic record of the heart's electrical activity?

A. Echocardiogram B. CT scan
C. Electrocardiogram (ECG) D. Angiogram

4. Which condition is most directly caused by atherosclerosis?

A. Anemia
B. Arrhythmia
C. Coronary artery disease
D. Deep vein thrombosis

5. What is the primary purpose of a coronary angiogram?

A. To check heart rhythm
B. To detect blood clots
C. To view blockages in coronary arteries
D. To measure blood viscosity

6. Which cardiovascular disorder is associated with irregular heartbeat?

A. Atherosclerosis B. Arrhythmia
C. Angina D. Myocardial infarction

7. Which of the following is a treatment for myocardial infarction?

A. Antihistamines B. Antibiotics
C. Thrombolytic therapy D. Antacids

8. Which of the following surgical procedures reroutes blood flow around blocked coronary arteries?

A. Angioplasty
B. Coronary bypass surgery
C. Valve replacement
D. Pacemaker implantation

Answer Key

1. B	2. C	3. C	4. C	5. C	6. B	7. C	8. B

Q.1 Briefly explain the purpose of an electrocardiogram (ECG).

Ans. An electrocardiogram (ECG) is a non-invasive test that records the heart's electrical activity over time. It detects and helps diagnose abnormalities in cardiac function by showing the depolarization and repolarization of the atria and ventricles. ECG is widely used to identify arrhythmias, which are irregular heart rhythms; myocardial infarction (heart attack), which results from blocked coronary arteries; ischemia, which indicates reduced blood flow to the heart; and other conditions affecting the cardiac conduction system. Clinicians rely on ECG for early detection, monitoring, and assessment of treatment outcomes, making it an essential diagnostic tool in cardiovascular medicine.

> **MARKING SCHEME (3 Marks)**
> - 1 mark each for any three cardiovascular disorders

Q.2 Describe angioplasty and its function in cardiovascular treatment.

Ans. Angioplasty is a minimally invasive medical procedure used to open narrowed or blocked coronary arteries, often caused by atherosclerosis. A catheter with a balloon tip is guided to the blockage site and inflated to widen the artery. In many cases, a stent is placed to maintain arterial patency and prevent re-narrowing. Angioplasty restores normal blood flow to the heart muscle, reducing symptoms such as chest pain (angina) and lowering the risk of heart attack. It is a life-saving intervention that improves cardiac function, enhances oxygen delivery to tissues, and is often combined with lifestyle modifications and medication to prevent recurrence of arterial blockage.

> **MARKING SCHEME (3 Marks)**
> - Definition – 1 mark
> - Function – 2 marks

Q.3 How does lifestyle influence the development and management of hypertension?

Ans. Lifestyle factors play a crucial role in both the development and control of hypertension (high blood pressure). Diets high in sodium and low in potassium can raise blood pressure, while excessive body weight increases cardiac workload. Physical inactivity, smoking, and chronic stress also contribute to elevated blood pressure and cardiovascular risk. Conversely, adopting healthy habits—such as reducing salt intake, engaging in regular aerobic exercise, maintaining a normal body weight, quitting smoking, and managing stress—can prevent or control hypertension. These measures improve vascular health, decrease strain on the heart, and enhance the effectiveness of antihypertensive medications, forming a cornerstone of non-pharmacological management.

> **MARKING SCHEME (3 Marks)**
> - 1 mark each for any three lifestyle factors with their impact

Q.4 What is coronary artery disease and how is it diagnosed?

Ans. Coronary artery disease (CAD) occurs when the coronary arteries supplying blood to the heart muscle become narrowed or blocked, usually due to atherosclerotic plaque buildup. Reduced blood flow can cause angina (chest pain), shortness of breath, or even myocardial infarction (heart attack). CAD is diagnosed using several clinical tools. An electrocardiogram (ECG) detects abnormal heart rhythms and signs of ischemia. Stress testing evaluates heart function under exertion. Coronary angiography visualizes arterial blockages directly, while blood lipid profiling assesses risk factors such as high cholesterol. Early diagnosis is essential to prevent complications and guide medical or interventional treatment.

> **MARKING SCHEME (3 Marks)**
> - Definition – 1 mark
> - Diagnostic techniques – 2 marks

Q.5 What is atherosclerosis and how does it impact cardiovascular health?

Ans. Atherosclerosis is a chronic condition in which fatty deposits, cholesterol, and cellular debris accumulate on the inner walls of arteries, forming plaques. These plaques narrow the arterial lumen, reducing blood flow and increasing vascular resistance. This restricts oxygen delivery to organs and tissues and can lead to complications such as coronary artery disease, stroke, or peripheral vascular disease. Plaque rupture may trigger blood clots, causing heart attacks or other acute events. Atherosclerosis also contributes to hypertension by stiffening arterial walls. Lifestyle factors, including diet, exercise, and smoking, play a significant role in both its development and prevention, highlighting the importance of early intervention and monitoring.

> **MARKING SCHEME (3 Marks)**
> - Definition – 1 mark
> - Effect/impact – 2 marks

Q.6 Define arrhythmia and mention one possible cause.

Ans. Arrhythmia refers to an irregularity in the heart's rhythm, which occurs when the electrical impulses that coordinate heartbeats are disrupted. This can manifest as a heart beating too fast, too slow, or with an irregular pattern. Arrhythmias may result from several factors, including electrolyte imbalances (e.g., abnormal potassium or calcium levels), ischemia due to reduced blood flow to the heart, side effects of certain medications, or damage to heart tissue following a myocardial infarction. Recognizing arrhythmias is important because they can compromise cardiac output and lead to dizziness, fainting, or even sudden cardiac death if untreated. Early detection through ECG and medical management is crucial.

MARKING SCHEME (3 Marks)
- Definition – 1 mark
- Cause – 2 marks

Q.7 What is an echocardiogram and what does it assess?

Ans. An echocardiogram is a non-invasive, ultrasound-based imaging technique that provides real-time visualization of the heart's structure and function. It allows clinicians to assess the size and movement of the heart chambers, evaluate valve structure and function, and measure the ejection fraction, which reflects the efficiency of the heart's pumping action. Echocardiography is commonly used to diagnose conditions such as heart valve disorders, cardiomyopathy, and heart failure, and it helps monitor disease progression and response to treatment. The test is safe, does not involve radiation, and offers crucial diagnostic information to guide clinical decision-making.

MARKING SCHEME (3 Marks)
- Definition – 1 mark
- Function – 2 marks

Q.8 Describe one non-surgical and one surgical treatment for blocked coronary arteries.

Ans. Non-surgical treatment: Angioplasty is a minimally invasive procedure in which a balloon-tipped catheter is guided to a narrowed coronary artery and inflated to widen the vessel. Often, a stent is placed to maintain patency and ensure continued blood flow to the heart muscle.
Surgical treatment: Coronary artery bypass graft (CABG) surgery involves using a grafted blood vessel, typically from the leg or chest, to bypass blocked or narrowed coronary arteries. This restores adequate perfusion to the myocardium, reducing the risk of heart attack and improving oxygen delivery. Both approaches aim to relieve symptoms and improve heart function.

MARKING SCHEME (3 Marks)
- Non-surgical treatment – 1.5 marks
- Surgical treatment – 1.5 marks

Q.9 What is heart failure and what are two common symptoms?

Ans. Heart failure is a chronic condition in which the heart is unable to pump blood efficiently to meet the body's metabolic demands. This leads to inadequate tissue perfusion and fluid accumulation in the lungs and peripheral tissues. Common symptoms include shortness of breath (dyspnea), especially during exertion or lying down, and fatigue due to reduced oxygen delivery to muscles. Other frequent signs are swelling (edema) in the legs and ankles and persistent coughing or wheezing caused by pulmonary congestion. Early detection and management, including lifestyle changes and medication, are essential to improve quality of life and reduce the risk of complications such as sudden cardiac arrest.

MARKING SCHEME (3 Marks)
- Definition – 1 mark
- Two common symptoms – 2 marks

Q.10 How can lifestyle modifications help prevent cardiovascular disorders?

Ans. Lifestyle modifications play a critical role in preventing cardiovascular diseases. Regular aerobic exercise strengthens the heart and improves blood circulation, while a balanced diet low in saturated fats, cholesterol, and sodium reduces the risk of atherosclerosis and hypertension. Maintaining a healthy weight decreases cardiac workload, and stress management helps control blood pressure and heart rate. Smoking cessation significantly lowers the risk of coronary artery disease, stroke, and peripheral vascular disease. Collectively, these habits improve vascular health, reduce plaque formation, and enhance overall cardiovascular efficiency, serving as an effective preventive strategy alongside medical interventions for at-risk individuals.

MARKING SCHEME (3 Marks)
- 1 mark each for any three lifestyle modifications

SLO [B-12-R-55]
Define hypertension and describe the factor that regulate blood pressure and lead to hypertension and hypotension.

HYPERTENSION AND HYPOTENSION

Hypertension

Hypertension is a condition characterised by persistently elevated arterial blood pressure above the normal range. Clinically, it is defined as a systolic pressure above 140 mmHg and/or a diastolic pressure above 90 mmHg. Hypertension increases the workload on the heart and arteries, making individuals more susceptible to cardiovascular complications such as heart attack, stroke, and kidney disease.

Factors Regulating Blood Pressure

Blood pressure is determined by the interaction of cardiac output, blood volume, and resistance in the blood vessels. Several physiological and external factors play a role:

1. **Cardiac Factors:**
   - **Heart Rate and Stroke Volume:** Increased heart rate or stronger ventricular contractions raise cardiac output, elevating blood pressure.
   - **Myocardial Contractility:** Enhanced contractility increases the force of blood ejection, contributing to higher pressure.
2. **Vascular Factors:**
   - **Peripheral Resistance:** Narrowing of arterioles (vasoconstriction) increases resistance, raising blood pressure.
   - **Elasticity of Arteries:** Stiff or hardened arteries cannot absorb the pressure effectively, leading to elevated systolic pressure.
3. **Blood Volume and Fluid Balance:**
   - **Renal Regulation:** Kidneys control blood volume through sodium and water retention or excretion. Excess fluid increases blood pressure, while fluid loss lowers it.
4. **Nervous System Regulation:**
   - **Autonomic Nervous System:** The sympathetic division increases heart rate and vasoconstriction (raising BP), while the parasympathetic division reduces heart rate and promotes vasodilation (lowering BP).
5. **Hormonal Factors:**
   - **Renin-Angiotensin-Aldosterone System (RAAS):** Stimulates vasoconstriction and sodium retention, raising BP.
   - **Antidiuretic Hormone (ADH):** Promotes water retention, increasing blood volume and pressure.
   - **Atrial Natriuretic Peptide (ANP):** Lowers blood pressure by promoting sodium and water excretion.

▸ Causes of Hypertension

- Genetic predisposition and family history.
- Obesity, high-salt diet, excessive alcohol intake, and sedentary lifestyle.
- Chronic stress and certain medications.
- **Secondary causes:** kidney disease, endocrine disorders, or arterial narrowing.

▸ Causes of Hypotension

- Excessive blood loss, dehydration, or fluid imbalance.
- Heart problems reducing cardiac output (e.g., heart failure, bradycardia).
- Hormonal deficiencies (e.g., Addison's disease).
- Severe infection (septic shock) or allergic reactions (anaphylaxis).

▸ Hypotension

Hypotension is defined as a condition in which **blood pressure falls below 90/60 mmHg**. It indicates insufficient force to maintain adequate perfusion of vital organs.

Symptoms of hypotension may include **dizziness, fainting, blurred vision, fatigue**, and in severe cases, **shock**. While low blood pressure may be benign in some individuals, it can be a sign of underlying pathology such as **cardiac failure, endocrine disorders**, or **severe dehydration**.

Feature	Hypertension	Hypotension
Definition	Abnormally high blood pressure	Abnormally low blood pressure
Typical Reading	≥140/90 mmHg	≤90/60 mmHg
Causes	Stress, obesity, high salt intake, genetics	Dehydration, blood loss, hormonal imbalance
Effects on Health	Increases risk of stroke, heart disease, kidney damage	May cause dizziness, fainting, shock
Symptoms	Often asymptomatic; may include headaches or blurred vision	Light-headedness, fatigue, fainting
Management	Lifestyle changes, antihypertensive drugs	Increased fluid and salt intake, treating underlying cause

SLO [B-12-R-56]
List the changes in lifestyles that can protect man from hypertension and cardiac problems.

LIFESTYLE CHANGES TO PREVENT HYPERTENSION AND CARDIAC PROBLEMS

Cardiovascular health is strongly influenced by lifestyle choices. Adopting healthy habits can significantly reduce the risk of **high blood pressure, heart attack, stroke, and other cardiac diseases.**

1. **Healthy Diet**
   - **Reduce Salt Intake:** Excess sodium increases blood pressure by retaining water. Aim for less than 5 g of salt per day.
   - **Balanced Nutrition:** Eat a variety of fruits, vegetables, whole grains, and lean proteins.
   - **Limit Saturated Fats and Trans Fats:** Reduces cholesterol buildup in arteries. Use healthy fats like olive oil or nuts.
   - **Moderate Sugar Consumption:** High sugar intake contributes to obesity and insulin resistance, which increase cardiac risk.
2. **Regular Physical Activity**
   - **Aerobic Exercise:** Activities like brisk walking, jogging, cycling, or swimming for at least 30 minutes daily improve heart efficiency and lower blood pressure.
   - **Strength Training:** Enhances muscle mass and metabolism, supporting overall cardiovascular health.
   - **Consistency:** Regular exercise maintains healthy body weight and reduces arterial stiffness.
3. **Maintain Healthy Body Weight**
   - **BMI and Waist Circumference:** Keeping a healthy body mass index (18.5–24.9) reduces strain on the heart and blood vessels.
   - **Fat Distribution:** Central obesity (excess abdominal fat) is a major risk factor for hypertension and heart disease.
4. **Avoid Smoking and Limit Alcohol**
   - **Smoking Cessation:** Nicotine damages arterial walls and promotes plaque formation.
   - **Alcohol Moderation:** Excessive alcohol raises blood pressure and contributes to heart failure.
5. **Stress Management**
   - ***Relaxation Techniques:*** Meditation, yoga, and deep-breathing exercises reduce sympathetic nervous system activity, lowering blood pressure.
   - **Time Management and Healthy Sleep:** Adequate rest (7–8 hours) helps maintain hormonal balance and cardiac function.

6. **Regular Health Check-ups**
- **Monitor Blood Pressure:** Early detection of hypertension allows timely intervention.
- **Lipid Profile and Glucose Monitoring:** Helps prevent atherosclerosis and diabetes, both of which increase cardiac risk.

7. **Avoid Sedentary Behaviour**
- **Frequent Movement:** Break long periods of sitting with short walks or stretching.
- **Active Lifestyle:** Gardening, stair climbing, or cycling to work helps maintain cardiovascular fitness.

Summary: By adopting a **heart-healthy lifestyle**, including diet, exercise, weight management, and stress control, humans can **prevent or delay hypertension and reduce the risk of cardiovascular diseases**, improving longevity and quality of life.

SLOs Based (MCQs)

HYPERTENSION AND HYPOTENSION

1. **Which of the following is a common symptom of hypotension?**
A. Chest pain B. Dizziness upon standing C. Severe headache D. Palpitations

2. **What is the normal adult blood pressure range considered by physicians?**
A. 140/90 mmHg B. 100/60 mmHg C. 120/80 mmHg D. 90/60 mmHg

3. **What is the primary function of baroreceptors in blood pressure regulation?**
A. Stimulate heart contraction B. Detect blood oxygen levels
C. Monitor arterial pressure D. Control red blood cell count

Answer Key

1. B	2. C	3. C

Short Questions

Q.1 Define hypertension and state its clinical significance.

Ans. Hypertension is a medical condition in which blood pressure remains consistently above 140/90 mmHg. It places excessive strain on the heart and blood vessels, increasing the risk of serious complications such as stroke, heart failure, kidney disease, and aneurysm formation. Because it often develops without noticeable symptoms, it is commonly called the "silent killer." Persistent high blood pressure accelerates arterial damage and promotes atherosclerosis, further heightening cardiovascular risk. Early detection and management through lifestyle interventions and medication are crucial to prevent long-term organ damage, improve quality of life, and reduce morbidity and mortality associated with hypertensive complications.

> **MARKING SCHEME (3 Marks)**
> - Definition – 1 mark
> - Clinical significance – 2 marks

Q.2 What are the main symptoms and consequences of hypotension?

Ans. Hypotension is defined as abnormally low blood pressure, typically below 90/60 mmHg. It can result in inadequate blood flow to vital organs. Common symptoms include dizziness, lightheadedness, fainting, fatigue, and blurred vision. In severe cases, hypotension can lead to shock, characterized by organ hypoperfusion and potentially life-threatening consequences. The brain and kidneys are particularly sensitive to reduced blood flow, which can cause confusion, acute kidney injury, or even organ failure. Early recognition of hypotension and its underlying causes is important for preventing complications and restoring normal circulation through fluid replacement, medication, or lifestyle adjustments.

> **MARKING SCHEME (3 Marks)**
> - Symptoms – 2 marks
> - Consequences/effects – 1 mark

Q.3 What role do baroreceptors play in cardiovascular regulation?

Ans. Baroreceptors are specialized stretch-sensitive mechanoreceptors located in the walls of the aortic arch and carotid sinus. They continuously monitor arterial pressure and detect any increases or decreases. When blood pressure rises, baroreceptors send signals to the medulla oblongata, triggering parasympathetic activity that slows

> **MARKING SCHEME (3 Marks)**
> - Definition/location – 1 mark
> - Function – 2 marks

heart rate and dilates blood vessels. Conversely, when blood pressure falls, they stimulate sympathetic responses, increasing heart rate and inducing vasoconstriction. This negative feedback system helps maintain stable blood pressure, ensuring adequate perfusion of vital organs. Baroreceptors are essential for short-term regulation of blood pressure, particularly during sudden changes in posture, activity, or blood volume.

SLO [B-12-R-57]
Describe the formation, composition and function of intercellular fluid.

INTERCELLULAR FLUID (ICF) – FORMATION, COMPOSITION, AND FUNCTION

Intercellular fluid, also known as **tissue fluid or interstitial fluid**, is the fluid that surrounds the cells in tissues. It forms an important component of the body's extracellular fluid and plays a key role in maintaining **cellular homeostasis**.

1. Formation of Intercellular Fluid

Intercellular fluid is formed primarily through **the process of filtration** from blood plasma in the capillaries.

Mechanism:

1. Blood pressure within the capillaries (hydrostatic pressure) forces water and small solutes out of the capillaries into the surrounding tissue spaces.
2. Large plasma proteins (like albumin) remain in the capillaries, creating an **osmotic gradient** that regulates water movement.
3. Excess interstitial fluid is eventually returned to the bloodstream via **lymphatic vessels,** ensuring fluid balance.

2. Composition of Intercellular Fluid

- **Water:** Forms the major component (about 90–95%), serving as a solvent.
- **Electrolytes:** Sodium (Na^+), potassium (K^+), calcium (Ca^{2+}), magnesium (Mg^{2+}), chloride (Cl^-), and bicarbonate (HCO_3^-) ions maintain osmotic balance and membrane potential.
- **Nutrients:** Glucose, amino acids, fatty acids, and oxygen diffuse from the blood to cells.
- **Metabolic Wastes:** Carbon dioxide, urea, and other by-products diffuse from cells into the intercellular fluid for removal.
- **Small Proteins:** Occasionally, small plasma proteins may be present in very low concentrations.

3. Functions of Intercellular Fluid

1. **Medium for Exchange:** Facilitates **diffusion of nutrients, gases, and wastes** between blood and cells.
2. **Cell Support and Protection:** Provides a **cushion around cells**, protecting them from mechanical shock.
3. **Maintains Tissue Homeostasis:** Regulates osmotic pressure and fluid balance within tissues.
4. **Transport of Hormones and Signalling Molecules:** Acts as a medium for hormones, growth factors, and other signalling substances to reach target cells.
5. **Waste Removal:** Carries cellular wastes to lymph and blood for excretion.

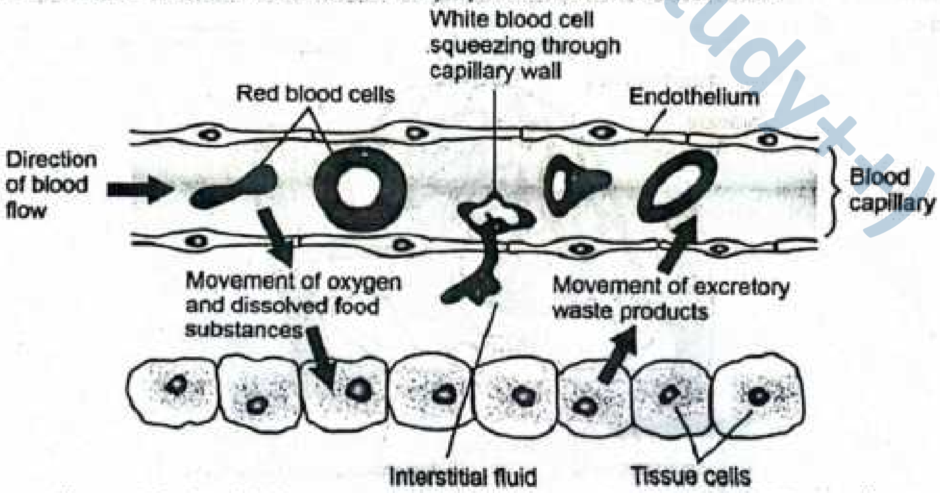


Figure: Relationship between a blood capillary, interstitial fluid and tissue cells

SLO [B-12-R-58]
Compare the composition of intercellular fluid with that of lymph.

COMPARISON OF INTERCELLULAR FLUID AND LYMPH

Intercellular fluid (ICF) and lymph are both extracellular fluids that play vital roles in **transporting nutrients, gases, and wastes** as well as in **immune surveillance**. While they are closely related, there are key differences in their composition and function.

Feature	Intercellular Fluid (ICF)	Lymph
Origin	Forms from blood plasma through capillary filtration.	Formed from intercellular fluid that enters lymphatic capillaries.
Appearance	Clear or slightly yellowish fluid.	Usually clear or pale yellow; may appear milky (chyle) after fat absorption in the small intestine.
Water Content	~90–95% water.	~95% water.
Electrolytes	Contains Na^+, K^+, Ca^{2+}, Mg^{2+}, Cl^-, HCO_3^- in concentrations similar to plasma but slightly lower protein content.	Similar electrolytes as plasma and intercellular fluid.
Proteins	Very low concentration; mostly small proteins that escape capillaries.	Higher protein content than intercellular fluid but lower than plasma; carries immune proteins (antibodies) in lymph nodes.
Nutrients	Glucose, amino acids, fatty acids, oxygen diffused from capillaries.	Nutrients transported from intercellular fluid; lymph from the small intestine (chyle) contains absorbed fats in addition.
Wastes	Carbon dioxide, urea, and metabolic by-products from cells.	Carries cellular wastes and excess fluid from tissues to the blood.
Special Components	Primarily acts as a medium for exchange between blood and cells.	Contains lymphocytes and immune cells; participates in immune surveillance.
Circulation	Mostly local diffusion around cells; returned to capillaries via osmotic gradients.	Flows through lymphatic vessels, lymph nodes, and eventually drains into the venous system.

- **Intercellular fluid** is the immediate environment around tissue cells, providing nutrients and removing wastes.
- **Lymph** is essentially interstitial fluid that has entered the lymphatic system; it carries additional proteins, fats, and immune cells, and eventually returns to the bloodstream.
- Both fluids are **interconnected**, with lymph acting as a conduit to maintain fluid balance and immune function.

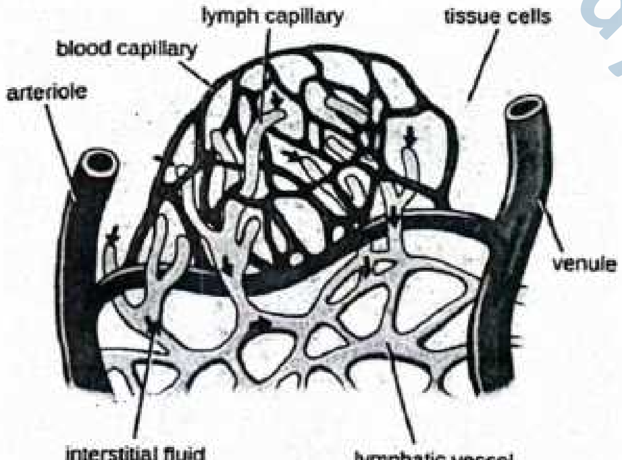


Figure: Lymphatic vessels

SLO [B-12-R-59]
State the structure and role of lymph capillaries, lymph vessels and lymph trunks.

7.

STRUCTURE AND ROLE OF LYMPHATIC VESSELS

The lymphatic system is a network of vessels that **collect, transport, and filter intercellular fluid (lymph)** before returning it to the bloodstream. It consists of **lymph capillaries, lymph vessels, and lymph trunks**, each with a specialized structure and role.

1. Lymph Capillaries

► **Structure:**

- **Smallest Lymphatic Vessels:** Lymph capillaries are microscopic and form the initial network of the lymphatic system. They are **blind-ended**, meaning one end is closed, allowing fluid to enter but not exit backward.
- **Thin Walls:** Composed of a **single layer of endothelial cells**, these walls are extremely delicate. The edges of these cells **overlap like shingles**, creating **one-way valves** that permit interstitial fluid, proteins, and small particles to enter but prevent backflow into the tissues.
- **Anchoring Filaments:** Fine filaments attach lymph capillaries to the surrounding connective tissue. These **prevent collapse** of the capillaries when interstitial pressure rises and **ensure continuous uptake** of intercellular fluid.
- **Permeability:** Lymph capillaries are highly permeable, allowing the absorption of fluids, macromolecules, and even fat droplets (chyle) from the intestinal villi during digestion.

► **Role:**

- **Collection of Intercellular Fluid:** Lymph capillaries absorb excess interstitial fluid that escapes from blood capillaries. This fluid includes water, dissolved salts, and small proteins, helping **maintain tissue fluid balance** and prevent edema.
- **One-Way Flow:** Thanks to the overlapping endothelial cells forming **one-way valves**, lymph capillaries allow fluid to enter while **preventing backflow,** ensuring unidirectional movement toward larger lymphatic vessels.
- **Transport of Fats (Chyle):** Specialized lymph capillaries in the small intestine, called **lacteals**, absorb dietary fats in the form of chylomicrons. These fats are transported via lymph to the **thoracic duct**, eventually entering the bloodstream.
- **Immune Surveillance:** By carrying proteins, debris, and foreign particles, lymph capillaries also help **transport antigens** to lymph nodes, initiating immune responses.

2. Lymph Vessels

► **Structure:**

- **Formation:** Lymph vessels are **larger channels** created by the merging of multiple lymph capillaries. This convergence allows the collected lymph to flow from tissues toward the central circulatory system.
- **Wall Composition:** Structurally similar to veins, lymph vessels have **three distinct layers:**
  (i) **Tunica intima:** Inner endothelial lining that provides a smooth surface for lymph flow.
  (ii) **Tunica media:** Middle layer containing smooth muscle fibers, which contract rhythmically to help propel lymph.
  (iii) **Tunica externa (adventitia):** Outer connective tissue layer providing support and flexibility.
- **Valves:** Lymph vessels contain **one-way valves at regular intervals**. These valves prevent backflow and ensure that lymph moves steadily toward the thoracic duct or right lymphatic duct, eventually entering the venous system near the heart.
- **Connectivity:** Lymph vessels connect **lymph capillaries to lymph trunks**, acting as intermediate channels that integrate local tissue drainage into systemic lymph circulation.

- **Role:**
  - **Lymph Transport:** Lymph vessels act as **conduits carrying lymph** from the tiny lymph capillaries toward larger lymph trunks and ducts. This ensures that excess interstitial fluid is returned to the circulatory system, maintaining **fluid balance**. 7
  - **Immune Cell Movement:** These vessels serve as **highways for immune cells**, such as lymphocytes and 7 macrophages, directing them toward **lymph nodes** where pathogens, debris, and antigens can be detected and processed.
  - **Protein and Lipid Transport:** Lymph vessels **carry proteins and fats** (especially chyle from the small intestine) that cannot be absorbed directly into blood capillaries. This contributes to **nutrient distribution and metabolic homeostasis**.
  - **Prevention of Edema:** By draining interstitial fluid efficiently, lymph vessels **prevent fluid accumulation in tissues**, which could otherwise lead to swelling or edema.
  - **Unidirectional Flow Maintenance:** The valves within lymph vessels ensure **one-way flow**, protecting tissues from backflow and enabling efficient lymph movement toward the **thoracic duct and right lymphatic duct**, where lymph rejoins the bloodstream.

3. Lymph Trunks

- **Structure:**
  - **Formation:** Lymph trunks are **large collecting vessels** formed by the convergence of smaller lymphatic vessels that drain lymph from major regions of the body.
  - **Wall Structure:** They are **larger and thicker-walled** compared to lymph vessels, with three distinct layers (tunica intima, tunica media, tunica externa), resembling veins in their structure.
  - **Valves:** Like other lymphatic vessels, lymph trunks contain **internal valves** to maintain unidirectional flow of lymph toward the thoracic and right lymphatic ducts.
  - **Regional Distribution:** Each lymph trunk drains lymph from a specific body region. Major lymph trunks include:
    - **Lumbar trunks** – drain lymph from lower limbs, pelvis, and parts of the abdominal wall.
    - **Intestinal trunk** – drains lymph rich in fats (chyle) from the gastrointestinal tract.
    - **Bronchomediastinal trunks** – drain lymph from thoracic organs and chest wall.
    - **Subclavian trunks** – drain lymph from the upper limbs and superficial thoracic wall.
    - **Jugular trunks** – drain lymph from the head and neck.
- **Role:**
  - **Regional Drainage:** Lymph trunks serve as major collecting channels, **draining lymph from large regions of the body** such as the limbs, head, thorax, abdomen, and gastrointestinal tract.
  - **Transport to Ducts:** They act as conduits, carrying lymph to the **two main lymphatic ducts**:
    - The **right lymphatic duct**, which drains lymph from the right upper quadrant of the body.
    - The **thoracic duct**, which drains lymph from the rest of the body.
    - Both ducts ultimately empty into the **subclavian veins**, where lymph rejoins the venous circulation.
  - **Maintaining Fluid Balance:** By returning excess intercellular fluid, proteins, and absorbed fats into the bloodstream, lymph trunks **help maintain blood volume and osmotic balance**, thereby preventing tissue swelling (edema).
  - **Supporting Immunity:** Through transport of antigen-presenting cells and lymphocytes toward lymph nodes and ducts, lymph trunks indirectly support the **body's immune defense mechanisms.**

Structure	Location/Structure Description	Function	Wall Structure
Lymphatic Capillaries	Microscopic, blind-ended vessels in tissue spaces	Absorb interstitial fluid and proteins to form lymph	Very thin, single layer of endothelium
Lymphatic Vessels	Larger vessels formed by convergence of capillaries	Transport lymph toward lymph nodes and larger trunks	Thinner than veins, contain valves
Lymphatic Trunks	Formed by union of several lymphatic vessels	Drain lymph from large body regions into collecting ducts	Thicker walls than vessels
Collecting Ducts	Thoracic duct and right lymphatic duct	Empty lymph into venous blood circulation (subclavian veins)	Large, thick-walled structures

SLO [B-12-R-60]
Describe the functions of lymph nodes and state the role of spleen as containing lymphoid tissue.

LYMPHATIC ORGANS AND THEIR FUNCTIONS

1. Lymph Nodes

Structure Overview:

- Small, bean-shaped organs located along the course of lymph vessels.
- Enclosed by a capsule and divided internally into a cortex (outer region) and medulla (inner region).
- Contain lymphocytes, macrophages, and reticular connective tissue.

Functions of Lymph Nodes:

- **Filtration of Lymph:** Trap and filter foreign particles, bacteria, and cellular debris from lymph before it re-enters the blood.
- **Immune Surveillance:** Lymphocytes (T-cells and B-cells) within the nodes detect antigens and initiate immune responses.
- **Phagocytosis:** Macrophages engulf and destroy pathogens and worn-out cells.
- **Lymphocyte Proliferation:** Provide sites where lymphocytes multiply and differentiate upon antigen exposure.
- **Barrier to Spread of Infection:** Prevent harmful microorganisms or cancer cells from spreading rapidly through the body.

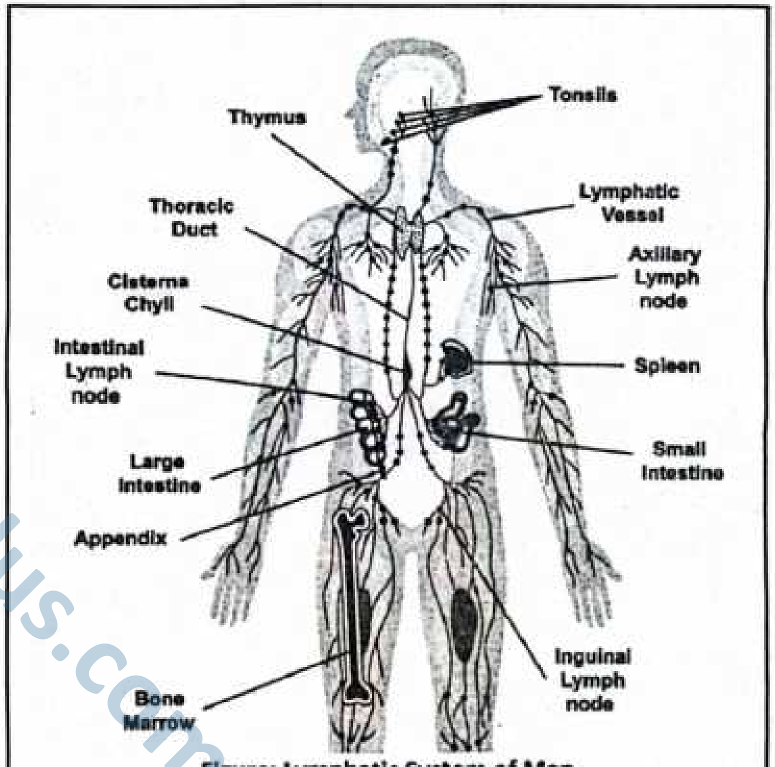


Figure: Lymphatic System of Man

2. Spleen (as Lymphoid Tissue)

Structure Overview:

- Largest lymphoid organ, located in the upper left abdomen behind the stomach.
- Composed of **red pulp** (rich in blood-filled sinuses and macrophages) and **white pulp** (clusters of lymphoid tissue with lymphocytes).

Role of Spleen:

- **Immune Function:**
  - White pulp contains lymphocytes that respond to antigens in the blood, producing antibodies.
  - Provides a site for immune surveillance and activation, similar to lymph nodes but for blood instead of lymph.
- **Blood Filtration:** Removes old or damaged red blood cells and platelets.
- **Storage:** Acts as a reservoir for blood, storing extra RBCs and platelets.
- **Phagocytosis:** Macrophages in red pulp destroy pathogens, cellular debris, and aged RBCs.
- **Support in Hematopoiesis:** In the fetus, the spleen contributes to blood cell formation; in adults, it may resume this function under certain disease conditions.

SLOs Based (MCQs)

LYMPHATIC SYSTEM

1. Which of the following structures absorbs fats from the digestive tract into the lymphatic system?

A. Lymph node B. Thoracic duct
C. Lacteal D. Spleen

2. The right lymphatic duct drains lymph from which of the following body regions?

A. Left arm and thorax
B. Right leg and abdomen
C. Right arm, right thorax, right side of head and neck
D. Entire lower body and left side of head

3. What feature of lymphatic vessels prevents the backflow of lymph?

A. Thick muscular walls
B. Bicuspid valves
C. Flap-like valves
D. High-pressure pumping

4. Which lymphatic organ acts as a reservoir for blood and also removes old red blood cells?

A. Lymph node B. Thymus
C. Spleen D. Tonsil

5. Compared to plasma, interstitial fluid has:

A. Higher protein content
B. Lower protein content
C. No electrolytes
D. More oxygen

Answer Key

1. C	2. C	3. C	4. C	5. B

Short Questions

Q.1 What is interstitial fluid and what is its role in the body?

Ans. Interstitial fluid, also called tissue fluid, is a clear fluid derived from blood plasma by capillary filtration. It surrounds body cells and fills the spaces between them. Its main role is to act as a medium for exchange between blood and tissues. Oxygen, nutrients, and hormones diffuse from the blood into the interstitial fluid before entering cells, while carbon dioxide and metabolic wastes move from cells into the fluid to return to the blood. By serving as an intermediary, it ensures proper cellular nutrition, waste removal, and maintenance of tissue homeostasis. Without interstitial fluid, cells would not receive adequate supplies nor efficiently eliminate wastes.

MARKING SCHEME (3 Marks)
- Definition – 1 mark
- Function – 2 marks

Q.2 Describe the structure and function of a lymphatic capillary.

Ans. Lymphatic capillaries are tiny, thin-walled vessels found in almost all tissues. Structurally, they are blind-ended tubes formed by a single layer of overlapping squamous epithelial cells. These cells act like one-way valves, allowing interstitial fluid to enter but preventing its escape. Once inside, the fluid is called lymph. Functionally, lymphatic capillaries collect excess interstitial fluid and return it to the bloodstream, thus preventing edema and maintaining fluid balance. Specialized lymphatic capillaries in the small intestine, called lacteals, also absorb dietary fats and transport them into the circulatory system. Thus, they play a dual role in fluid homeostasis and fat absorption.

MARKING SCHEME (3 Marks)
- Structure – 1.5 marks
- Function – 1.5 marks

Q.3 Differentiate between interstitial fluid and lymph.

Ans. Interstitial fluid is the plasma-derived liquid found in spaces between body cells. It delivers oxygen and nutrients to cells and collects waste products. Once this fluid enters lymphatic capillaries, it is termed lymph. Although both fluids are similar in composition, lymph differs because it contains higher levels of proteins and white blood cells, particularly lymphocytes. As lymph passes through lymph nodes, it becomes enriched with immune cells that provide defense against pathogens. In short, interstitial fluid primarily serves as a medium of exchange around tissues, while lymph represents this fluid after it enters lymphatic vessels, contributing actively to immune surveillance.

MARKING SCHEME (3 Marks)
- Properties of interstitial fluid – 1.5 marks
- Properties of lymph – 1.5 marks

Q.4 State the functions of lymph nodes.

Ans. Lymph nodes are small, bean-shaped structures located along lymphatic vessels. They act as biological filters for lymph before it re-enters the bloodstream. Their functions include: (1) trapping and destroying pathogens through lymphocytes and macrophages, (2) filtering out debris and damaged cells, and (3) initiating immune responses by activating lymphocytes against infectious agents. In this way, lymph nodes serve as checkpoints that prevent the spread of infection and contribute to overall body defense. Swelling of lymph nodes is often a clinical sign of infection, as immune cells multiply and respond to invading microorganisms.

> **MARKING SCHEME (3 Marks)**
> - Any three correct functions – 1 mark each

Q.5 What are the collecting ducts in the lymphatic system and what regions do they drain?

Ans. The lymphatic system has two main collecting ducts. The thoracic duct, the larger of the two, begins in the abdomen at the cisterna chyli and ascends through the thorax to drain into the left subclavian vein. It collects lymph from the entire lower body, the left arm, the left side of the thorax, and the left side of the head and neck. The right lymphatic duct is smaller and drains into the right subclavian vein. It collects lymph from the right arm, the right side of the thorax, and the right side of the head and neck. Together, these ducts ensure that lymph from the whole body is returned to the bloodstream.

> **MARKING SCHEME (3 Marks)**
> - Names of collecting ducts – 2 marks
> - Pathway/regions drained – 1 mark

Q.6 List two major immune functions of the spleen.

Ans. The spleen is a lymphoid organ with important roles in blood filtration and immunity. First, it produces lymphocytes and antibodies that help fight infections, strengthening both cellular and humoral immunity. Second, it contains macrophages that engulf and destroy bacteria, viruses, and worn-out blood cells, thereby purifying the blood. By removing pathogens and cellular debris, the spleen prevents infections from spreading and contributes to long-term immune memory. Additionally, its filtering action ensures that old or damaged red blood cells are removed, while useful components like iron are recycled. These immune functions make the spleen an essential defense organ.

> **MARKING SCHEME (3 Marks)**
> - Two immune functions – 1.5 marks each

Exercise

I Multiple Choice Questions (MCQs)

❖ **Select the Correct Answer.**

1. **Which one of the following has the thickest wall?**
   A. right ventricle B. left ventricle
   C. right atrium D. left atrium

2. **The reason why tricuspid and bicuspid valves are closed is**
   A. ventricular relaxation
   B. ventricular filling
   C. atrial systole
   D. attempted backflow of blood into the atria

3. **Tricuspid valve is present between**
   A. ventricle and pulmonary artery
   B. ventricle and aorta
   C. left auricle and left ventricle
   D. right auricle and right ventricle

4. **From where does the aorta originate?**
   A. Right ventricle B. Left atrium
   C. Right atrium D. Left ventricle

5. **How many aortic valves are present?**
   A. 1 B. 2
   C. 3 D. 4

6. **The epicardium:**
   A. Is also known as the parietal pericardium.
   B. Is a layer of cardiac muscle.
   C. Is the visceral pericardium.
   D. Lines the heart chambers.

7. **The tissue which forms a loose-fitting sac around the heart is the:**
   A. Visceral pericardium B. Parietal pericardium
   C. Myocardium D. Epicardium

8. **The valve between the left ventricle and the blood vessel leaving the left ventricle is the:**

A. Bicuspid valve
B. Tricuspid valve
C. Pulmonary semilunar valve
D. Aortic semilunar valve

9. During the period of ejection in the cardiac cycle, the atrioventricular valves are _______ and the semilunar valves are ________.
A. closed, closed B. closed, open
C. open, closed D. open, open

10. The second heart sound, described as "dupp" is actually the sound of the
A. Atria contracting
B. Ventricles contracting
C. Atrioventricular valves closing
D. Semilunar valves closing

11. Cardiac output is determined by:
A. heart rate B. stroke volume
C. blood flow
D. heart rate and stroke volume

12. Which one is the definition of cardiac cycle?
A. The contraction of the atria.
B. Circulation of blood in the heart.
C. The contraction and relaxation of the ventricles.
D. It is sequence of events that occurs during a complete heartbeat.

13. For how long is the cardiac cycle, if the heart rate is 75 beats /min ?
A. 0.7 sec B. 0.8 sec
C. 0.5 sec D. 0.4 sec

14. The fluid that passes through the lymphatic vessels
A. flows toward the lungs
B. passes from the lymphatic vessels into the arteries
C. enters the left ventricle of the heart through the right thoracic duct
D. moves in a single direction toward the heart

15. Lymph nodes may be located in the human body in the tissues of the
A. stomach and brain
B. groin and neck
C. ventricle and atrium
D. thyroid gland and adrenal gland

16. A red blood cell, entering the right side of the heart, passes by or through the following structures:
1. atrioventricular valve 2. semilunar valve 3. right atrium 4. right ventricle 5. Pulmonary trunk
In which order will the red blood cell passes the structures?
A. 2→3→4→5 B. 3→1→4→5
C. 3→5→1→4 D. 5→3→1→4→2

17. The rhythmic beating of cardiac muscle in the mammalian heart is initiated by the.
A. atrio-ventricular node
B. parasympathetic nervous system
C. Purkinje tissue
D. sino-atrial node

18. What produces systolic blood pressure?
A. contraction of the right atrium
B. contraction of the right ventricle
C. contraction of the left atrium
D. contraction of the left ventricle

19. Human heart is
A. myogenic B. neurogenic
C. cardiogenic D. digenic

20. Pacemaker is situated in heart
A. in the wall of right atrium
B. on interauricular septum
C. on interventricular septum
D. in the wall of left atrium

Answer Key

1. B	2. D	3. D	4. D	5. C	6. C	7. B	8. D	9. B	10. D
11. D	12. B	13. B	14. D	15. B	16. D	17. D	18. D	19. A	20. A

II Short Answer Questions

Q.1 *Why do we have a circulatory system?*

Ans. The circulatory system ensures efficient transport of materials throughout the body. It delivers oxygen, nutrients, and hormones to tissues while removing carbon dioxide and waste products. It also regulates body temperature, maintains homeostasis, and supports immunity by circulating white blood cells and antibodies. Diffusion alone would be too slow to meet the metabolic demands of complex, multicellular organisms, making a circulatory system essential for survival.

MARKING SCHEME (3 Marks)
- Transport role – 1 mark
- Regulation/homeostasis – 1 mark
- Limitation of diffusion – 1 mark

Q.2 What are the contraction and relaxation of the human heart called?

Ans. The contraction phase of the heart is called **systole**, when blood is pumped out of the atria and ventricles. The relaxation phase is called **diastole**, when the chambers expand and refill with blood. Together, systole and diastole form the **cardiac cycle**, which ensures continuous circulation of blood.

> **MARKING SCHEME (3 Marks)**
> - Systole – 1 mark
> - Diastole – 1 mark
> - Cardiac cycle link – 1 mark

Q.3 Where are the SA node, AV node, Bundle of His, and Purkinje fibres located?

Ans.
- **SA node:** wall of the right atrium, near opening of the superior vena cava.
- **AV node:** lower part of interatrial septum.
- **Bundle of His:** begins at AV node, extends into interventricular septum.
- **Purkinje fibres:** spread along inner ventricular walls.
- These structures form the heart's conduction system, ensuring coordinated contraction.

> **MARKING SCHEME (3 Marks)**
> - Any two correct locations – 1 mark each
> - Remaining locations – 0.5 mark each

Q.4 Why do action potentials travel along Purkinje fibres more rapidly than through other muscle fibres?

Ans. Purkinje fibres are specialised cardiac muscle fibres adapted for rapid conduction. They are wider in diameter and have fewer contractile myofibrils, which lowers resistance to impulse transmission. In addition, they contain abundant gap junctions that facilitate fast electrical communication between cells. These features allow impulses to spread quickly, ensuring that both ventricles contract almost simultaneously for efficient pumping.

> **MARKING SCHEME (3 Marks)**
> - Structural specialisation (large diameter, fewer myofibrils) – 1.5 marks
> - Functional outcome (rapid transmission, synchronised contraction) – 1.5 marks

Q.5 Name the artery supplying blood to the heart.

Ans. The heart receives oxygenated blood through the **coronary arteries**, which branch from the base of the aorta. These arteries spread across the surface of the myocardium to supply cardiac muscle. If a coronary artery becomes blocked, blood supply is reduced, leading to oxygen deprivation and possible **myocardial infarction (heart attack)**.

> **MARKING SCHEME (3 Marks)**
> - Name of coronary arteries – 1 mark
> - Origin (aorta) – 1 mark
> - Clinical significance (blockage effect) – 1 mark

Q.6 What is blood pressure?

Ans. Blood pressure is the force that circulating blood exerts against the walls of arteries as the heart pumps. It reflects how hard the heart is working and the resistance offered by blood vessels. Blood pressure is measured in millimetres of mercury (mmHg) and expressed as two readings: systolic pressure (when ventricles contract) over diastolic pressure (when ventricles relax). A normal adult reading is about 120/80 mmHg. Systolic pressure represents the maximum force during contraction, while diastolic indicates resting pressure. Blood pressure is essential for maintaining continuous blood flow to organs and tissues. Abnormally high blood pressure is called hypertension, while abnormally low pressure is hypotension, both of which can lead to serious health problems.

> **MARKING SCHEME (3 Marks)**
> - Definition – 1 mark
> - Measurement and normal value – 1 mark
> - Clinical note (hypertension / hypotension) – 1 mark

Q.7 Why is SA node called the pacemaker of the heart?

Ans. The sinoatrial (SA) node, located in the right atrium near the opening of the superior vena cava, plays a crucial role in regulating heartbeat. It contains specialised muscle cells capable of generating electrical impulses spontaneously without external stimulation. These impulses spread through the atrial walls, causing atrial contraction, and then reach the AV node, initiating ventricular contraction. Because it initiates each heartbeat and sets the pace at which the heart beats, the SA node is known as the natural pacemaker of the heart. Its normal rhythm maintains about 70–75 beats per minute in adults. If the SA node fails, artificial pacemakers may be implanted to maintain proper heart rhythm and cardiac function.

> **MARKING SCHEME (3 Marks)**
> - Location of SA node – 1 mark
> - Role in initiating impulses – 1 mark
> - Pacemaker significance – 1 mark

Q.8 What is a cardiac cycle?

Ans. A cardiac cycle is the complete sequence of events that occurs during one heartbeat. It consists of atrial systole (contraction of atria), ventricular systole (contraction of ventricles), and a relaxation phase called diastole. During this cycle, blood is pumped from the atria to the ventricles and then from the ventricles to the lungs and body. At a resting heart rate of about 75 beats per minute, each cardiac cycle lasts approximately 0.8 seconds: 0.1 sec for atrial systole, 0.3 sec for ventricular systole, and 0.4 sec for complete diastole. This cycle ensures continuous circulation of blood, providing oxygen and nutrients to tissues while removing wastes. Disturbance in the cardiac cycle can lead to arrhythmias and circulatory problems.

> **MARKING SCHEME (3 Marks)**
> - Definition – 1 mark
> - Phases with duration – 1 mark
> - Significance – 1 mark

Q.9 What is an arterial pulse? What is the normal human pulse rate?

Ans. The arterial pulse is the rhythmic expansion and recoil of arterial walls produced by the ejection of blood from the left ventricle during systole. Each pulse wave corresponds to one heartbeat and can be felt in superficial arteries such as the radial artery at the wrist or the carotid artery in the neck. The pulse provides a simple, non-invasive way to assess heart rate and rhythm. In healthy adults at rest, the normal pulse rate ranges between 70–75 beats per minute, though it may be higher in children and lower in trained athletes. Abnormal pulse rates may indicate heart disease, circulatory disorders, or underlying medical conditions that affect cardiovascular function.

> **MARKING SCHEME (3 Marks)**
> - Definition of arterial pulse – 1 mark
> - Pulse sites and measurement – 1 mark
> - Normal value – 1 mark

Q.10 Why is AV node essential for the conduction of cardiac impulse?

Ans. The atrioventricular (AV) node is a small cluster of specialised cells located in the lower interatrial septum near the tricuspid valve. It plays a vital role in coordinating heart contractions by acting as a relay station between the atria and ventricles. When the SA node generates an impulse, the AV node delays its transmission for about 0.1 seconds. This short delay allows the atria to contract fully and push blood into the ventricles before they contract. After the delay, the impulse passes through the Bundle of His, bundle branches, and Purkinje fibres to trigger ventricular contraction. Without this delay, atria and ventricles would contract simultaneously, reducing pumping efficiency. Thus, the AV node ensures proper cardiac rhythm and circulation.

> **MARKING SCHEME (3 Marks)**
> - Location of AV node – 1 mark
> - Delay function – 1 mark
> - Importance in coordination – 1 mark

Q.11 What are the risks associated with atherosclerosis?

Ans. Atherosclerosis is a serious cardiovascular condition caused by the buildup of fatty deposits, cholesterol, and fibrous tissue (plaques) inside arterial walls. This narrows and hardens the arteries, reducing blood flow to vital organs. As a result, it significantly increases the risk of **coronary artery disease**, where the heart muscle receives insufficient oxygen. It may also cause **myocardial infarction (heart attack)** if a plaque ruptures and blocks a coronary artery. In the brain, it can lead to **stroke**, while in peripheral arteries it results in **peripheral artery disease** causing pain and tissue damage. Additionally, atherosclerosis contributes to **high blood pressure** and arterial rupture, making it one of the leading causes of morbidity and mortality worldwide.

> **MARKING SCHEME (3 Marks)**
> - 3 points 1 mark each (heart, brain, peripheral vessels/other complications).

Q.12 Why can you feel your pulse in arteries but not in veins? If there is no pulse in your veins, what pushes the blood in veins back to the heart?

Ans. A **pulse** can be felt in arteries because they are directly connected to the pumping action of the heart. Each time the ventricles contract (systole), a **pressure wave** travels along the arteries, causing them to expand and recoil. This rhythmic expansion produces the arterial pulse, which is absent in veins because venous blood flows at **much lower pressure.** Veins, therefore, lack pulsation. Instead, venous return to the heart depends on other mechanisms. The most important are the **skeletal muscle pump**, in which contracting muscles squeeze veins, and the **presence of one-way valves**, which prevent backflow. Additionally, **respiratory movements** create pressure changes that assist the upward flow of venous blood toward the right atrium.

> **MARKING SCHEME (3 Marks)**
> - 1 mark for arteries/pulse explanation, 1 mark for veins/no pulse, 1 mark for mechanisms of venous return.

Q.13 Define the term thrombus and differentiate between thrombus and embolus.

Ans. A **thrombus** is a solid mass of blood components (platelets, fibrin, and trapped red blood cells) that forms abnormally inside a blood vessel. It remains **fixed at the site of origin** and may obstruct blood flow locally, leading to conditions like deep vein thrombosis or coronary artery blockage. An **embolus**, on the other hand, refers to a detached thrombus fragment, air bubble, fat droplet, or any foreign substance traveling freely in the bloodstream. When it lodges in a narrow vessel, it may block circulation at a site **different from its origin**, causing complications such as pulmonary embolism or stroke. Thus, while both are dangerous, the key difference lies in whether the clot is **stationary (thrombus)** or **mobile (embolus)**.

> **MARKING SCHEME (3 Marks)**
> - Definition of thrombus – 1 mark
> - Definition of embolus – 1 mark
> - Difference (stationary vs mobile) – 1 mark.

Q.14 Identify the factors causing atherosclerosis and arteriosclerosis.

Ans. Several risk factors contribute to the development of **atherosclerosis** (plaque buildup in arteries) and **arteriosclerosis** (general thickening and hardening of arterial walls). Major lifestyle-related causes include **high intake of saturated fats and cholesterol, smoking, sedentary habits,** and **obesity**, all of which damage arterial endothelium or raise blood lipid levels. Medical conditions such as **hypertension (high blood pressure)** and **diabetes mellitus** further accelerate the process by stressing or damaging arterial linings. Atherosclerosis specifically involves **fatty plaques** narrowing the lumen, whereas arteriosclerosis may occur with or without plaque deposition, often associated with aging. Genetic predisposition, stress, and excessive alcohol intake also increase susceptibility. Preventive measures such as a healthy diet, regular exercise, and blood pressure control are crucial.

> **MARKING SCHEME (3 Marks)**
> - At least 3 factors – 1 mark each
> - Distinction between atherosclerosis and arteriosclerosis – 1 mark.

Q.15 List the advantages and disadvantages of coronary bypass.

Ans. **Coronary artery bypass grafting (CABG)** is a surgical procedure that restores blood flow to the heart muscle by rerouting blood around blocked coronary arteries. **Advantages include** improved blood supply to the myocardium, significant relief from **angina (chest pain)**, reduced risk of future **heart attacks**, and improvement in overall quality of life. It allows patients to regain normal physical activity and extends survival in severe coronary artery disease. However, there are **disadvantages** as well. Being a major operation, CABG carries risks of **infection, bleeding, or complications from anaesthesia**. Recovery is long, often requiring weeks of rest and rehabilitation. Patients must also adopt permanent **lifestyle changes** such as a healthy diet, exercise, and quitting smoking to maintain long-term benefits.

> **MARKING SCHEME (3 Marks)**
> - Advantages – 1.5 marks (any 2 points), Disadvantages – 1.5 marks (any 2 points).

Q.16 List the changes in lifestyle that can protect man from hypertension and cardiac problems.

Ans. Lifestyle modification is the most effective way to reduce the risk of hypertension and cardiovascular disease. A balanced diet rich in fruits, vegetables, whole grains, and lean proteins helps maintain healthy blood pressure and cholesterol levels. Limiting intake of salt, saturated fats, and processed foods prevents arterial damage. Regular aerobic exercise such as walking, jogging, or swimming strengthens the heart and improves circulation. Avoiding smoking and limiting alcohol intake reduce stress on blood vessels. Maintaining a healthy body weight decreases the workload on the heart, while effective stress management through relaxation, meditation, or adequate sleep helps stabilize blood pressure. Together, these lifestyle changes provide long-term protection against hypertension and cardiac complications.

> **MARKING SCHEME (4 Marks)**
> - Any four lifestyle changes – 1 mark each (total 4 marks)

Q.17 What is the major feature of human lymphatic system?

Ans. The lymphatic system is a parallel circulatory network that complements the blood vascular system. Its major feature is the collection of excess interstitial fluid, which is drained into lymphatic capillaries and transported as lymph back into the venous blood. This prevents tissue swelling (edema) and maintains fluid balance. Lymph nodes distributed along the vessels act as biological filters that trap pathogens, foreign particles, and abnormal cells, thereby supporting immune defence. The system also plays a vital role in absorbing dietary fats through specialized lymphatic vessels in the intestinal villi known as

> **MARKING SCHEME (3 Marks)**
> - Fluid return – 1 mark
> - Immune function – 1 mark
> - Fat absorption – 1 mark
> - Transport of lymphocytes – 1 mark

lacteals. In addition, it provides a route for the circulation of lymphocytes, ensuring continuous immune surveillance throughout the body.

Q.18 Justify why blood circulatory system is dependent on the lymphatic system.

Ans. The blood circulatory system and lymphatic system function together to maintain homeostasis. During normal blood circulation, a portion of plasma fluid leaks from capillaries into the surrounding tissues. Without the lymphatic system, this excess fluid would accumulate, causing edema and impairing circulation. Lymphatic vessels collect this fluid along with plasma proteins and return them to the venous system, thereby conserving blood volume and maintaining normal blood pressure. Additionally, lymph nodes filter lymph and trap harmful microbes or cancer cells, providing protection to the bloodstream. The lymphatic system also facilitates the transport of lymphocytes, ensuring immune surveillance. Thus, the cardiovascular system depends on the lymphatic system for fluid balance, pathogen defence, and overall efficiency.

MARKING SCHEME (3 Marks)
- Fluid return – 1 mark
- Protein return – 1 mark
- Immune defence – 1 mark
- Justification of interdependence – 1 mark

Q.19 Interpret why the swelling of the lymph node is cause of concern.

Ans. Swelling of a lymph node, medically known as lymphadenopathy, is a clinical sign that the body is responding to an abnormal condition. It often occurs when lymph nodes actively filter pathogens such as bacteria or viruses during infection, leading to enlargement due to increased lymphocyte activity. Inflammatory conditions like autoimmune disorders may also cause persistent swelling. More seriously, swollen nodes can indicate malignant changes such as lymphoma or the spread of cancer cells from nearby tissues. Because of these possibilities, lymph node swelling is always significant. Temporary swelling during mild infections is common, but persistent or painless enlargement requires medical evaluation to rule out serious diseases, making it an important diagnostic clue.

MARKING SCHEME (3 Marks)
- Infection/inflammation cause – 1 mark
- Immune activity explained – 1 mark
- Malignancy link – 1 mark
- Diagnostic concern – 1 mark

Q.20 Write the differences between:

(a) bicuspid valve and tricuspid valve.

Ans. The **bicuspid valve**, also called the **mitral valve**, is located between the **left atrium and left ventricle**. It has **two cusps (leaflets)** that prevent backflow of blood into the left atrium during ventricular contraction. Since the left ventricle pumps blood into the systemic circulation, the bicuspid valve must withstand higher pressure. In contrast, the **tricuspid valve** lies between the **right atrium and right ventricle**. It has **three cusps**, which stop blood from flowing back into the right atrium during contraction of the right ventricle. This valve operates under relatively lower pressure, as the right ventricle pumps blood only to the lungs. Both valves ensure **unidirectional blood flow,** maintaining efficiency of the cardiac cycle.

MARKING SCHEME (3 Marks)
- Location – 1 mark
- Structure (cusps) – 1 mark
- Functional difference – 1 mark

(b) Systole and Diastole

Ans. **Systole** refers to the **contraction phase** of the cardiac cycle. During atrial systole, atria contract to push blood into ventricles, while during *ventricular* systole, the ventricles contract forcefully to pump blood into the pulmonary artery and aorta. This phase is associated with **higher pressure** and the opening of semilunar valves. **Diastole**, in contrast, is the **relaxation phase** of the heart. Here, chambers relax and fill with blood: atria fill from veins, and ventricles fill from atria. Diastole allows proper circulation by ensuring chambers are refilled before the next *contraction*. Together, systole and diastole complete one heartbeat, lasting about **0.8 seconds** at rest. Their balance is essential for maintaining normal blood pressure and effective circulation.

MARKING SCHEME (3 Marks)
- Definition of systole – 1 mark
- Definition of diastole – 1 mark
- Functional difference – 1 mark

(c) SA node and AV node

Ans. The **SA node (sinoatrial node)** is located in the wall of the right atrium **near the superior vena cava opening.** It initiates electrical impulses that stimulate atrial contraction and sets the pace of the heartbeat, earning the title **"natural pacemaker"** of the heart. The **AV node (atrioventricular node),** on the other hand, lies in the **interatrial septum** near the tricuspid valve. It receives impulses from the SA node,

MARKING SCHEME (3 Marks)
- Location difference – 1 mark
- Role of SA node – 1 mark
- Role of AV node – 1 mark

introduces a slight **delay**, and then passes the signal to the Bundle of His and Purkinje fibres, leading to ventricular contraction. This delay ensures atria empty blood fully into ventricles before they contract. Thus, SA node controls rhythm, while AV node coordinates timing.

(d) P-wave and T-wave of ECG

Ans. In an **ECG (electrocardiogram)**, the **P-wave** represents **atrial depolarisation**, which is the electrical activity causing contraction of the atria. It occurs at the beginning of the cardiac cycle and ensures blood is pushed into the ventricles. The **T-wave**, in contrast, reflects **ventricular repolarisation**, which corresponds to relaxation of the ventricles after contraction. While the P-wave is generally smaller in amplitude and occurs early in the cycle, the T-wave is broader and appears later. Abnormalities in P-wave may indicate atrial enlargement or conduction problems, while irregular T-waves may suggest ischemia, electrolyte imbalance, or other ventricular issues. Together, these waves are critical for diagnosing cardiac health and rhythm disorders.

MARKING SCHEME (3 Marks)
- P-wave description – 1 mark
- T-wave description – 1 mark
- Timing/functional difference – 1 mark

(e) Blood capillaries and Lymph capillaries

Ans. **Blood capillaries** are fine, thin-walled vessels formed of a single layer of endothelial cells that connect arterioles with venules. They allow exchange of oxygen, carbon dioxide, nutrients, and wastes between blood and tissues. Blood within them flows in a **continuous circuit** under pressure. **Lymph capillaries**, however, are **blind-ended vessels** present in tissue spaces. They absorb excess interstitial fluid, which then becomes lymph, and transport it to larger lymphatic vessels. They are more permeable than blood capillaries and also absorb **dietary fats** in the intestine through structures called lacteals. Unlike blood capillaries, lymph capillaries carry fluid in **one direction only**, towards the heart, maintaining tissue fluid balance.

MARKING SCHEME (3 Marks)
- Structure (continuous vs blind-ended) – 1 mark
- Function (exchange vs fluid return) – 1 mark
- Direction/permeability difference – 1 mark

(f) Baroreceptor and Volume Receptor

Ans. **Baroreceptors** are specialised stretch-sensitive receptors located mainly in the **carotid sinus and aortic arch**. They detect changes in **arterial blood pressure** and send signals to the medulla to adjust heart rate and vascular tone, thus helping to maintain blood pressure homeostasis. In contrast, **volume receptors** (also called atrial receptors) are found in the **walls of atria and pulmonary veins**. They respond to changes in **blood volume**, especially venous return and atrial filling. Their stimulation influences kidney function and hormone secretion (e.g., ADH release), regulating long-term blood volume. Thus, baroreceptors mainly regulate **short-term pressure control**, while volume receptors *regulate* **long-term fluid balance** in the circulatory system.

MARKING SCHEME (3 Marks)
- Baroreceptor role – 1 mark
- Volume receptor role – 1 mark
- Difference in function/response – 1 mark

III Long Answer Questions

Q.1 Draw, label and describe the external structure of human heart.

Ans. Please See SLO [B-12-R-37]

Q.2 Describe the flow of blood through human heart as regulated by the valves.

Ans. Please See SLO [B-12-R-38]

Q.3 State the phases of heartbeat in man.

Ans. Please See SLO *[B-12-R-39]*

Q.4 Describe the conducting system of human heart.

Ans. Please See SLO [B-12-R-40]

Q.5 Explain electrocardiogram with the help of diagram.

Ans. Please See SLO [B-12-R-41]

Q.6 Describe the structure of blood vessels in man.

Ans. Please See SLO [B-12-R-42]

Q.7 What is the role of precapillary sphincter?

Ans. Please See SLO [B-12-R-44]

Q.8 Describe pulmonary circulation and systemic circulation.

Ans. Please See SLO [B-12-R-45]

Q.9 Describe hepatic portal system.

Ans. Please See SLO [B-12-R-45]

Q.10 Give an account of blood pressure in ma...

Ans. Please See SLO [B-12-R-47]

Q.11 Compare the rate of blood flow through arteries, arterioles, capillaries, venules and veins.

Ans. Please See SLO [B-12-R-46]

Q.12 Explain the following:
a. Principle of angiography b. Coronary bypass c. Angioplasty d. Open heart surgery

Ans. Please See SLO [B-12-R-54]

Q.13 Explain hypertension and hypotension. What are the factors that regulate blood pressure?

Ans. Please See SLO [B-12-R-55]

Q.14 Describe the lymphatic system of man.

Ans. Please See SLO [B-12-R-59] and [B-12-R-60]